

A cut-cell immersed boundary method for static and moving bodies

with passive scalar transport in a collocated finite-volume solver

Formulation, implementation in OpenFOAM and verification

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Abstract

We present a cut-cell immersed boundary method for the incompressible Navier–Stokes equations on a fixed, non body-fitted mesh, its extension to rigid bodies moving through the mesh, and the transport of passive scalars in the same framework. The body is described by a level set; each face of the mesh carries the fraction of its area that lies in the fluid and each cell the fraction of its volume. The wall segment of a cut cell is not computed independently but *defined* by the geometric closure of the fluid polyhedron, and this single identity carries the whole boundary treatment: the wall pressure force is absorbed into a face sum, no-penetration follows from discrete continuity, and the difference between a free-slip and a no-slip wall is one diagonal coefficient. No ghost values, no extrapolation into the solid and no coupling between velocity components appear. For a moving body only continuity changes: it acquires the flux swept by the wall as a source, and taking that flux — rather than the geometric volume change — as the primary quantity makes the discrete geometric conservation law hold by construction, so that a free stream is preserved to machine precision at any body speed. Passive scalars are transported by the same face fluxes and see the wall either as a fixed-value or as a zero-flux surface; their budgets close to round-off. An optional second-order wall treatment replaces the cell centres by the centroids of the fluid parts, with a centroid-consistent diffusive face flux and a centroid wall distance; it raises the operators and the velocity from first to second order without ghost values or stencil growth. The method is implemented in OpenFOAM without modifying the mesh: the fractions enter the standard operators explicitly, which keeps the pressure–velocity coupling of `simpleFoam`/`pimpleFoam` and the linear algebra untouched, and the whole moving-body extension sits behind a single run-time switch. The implementation is verified against an independent reference implementation of the same discretisation on a cylinder at $Re_D = 1$ at rest, oscillating and translating through a periodic seam, by conservation tests for the scalar, and by order-of-accuracy studies on manufactured and exact solutions.

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1 Introduction

Immersed boundary methods let a body of arbitrary shape be represented on a mesh that does not conform to it. Among them, cut-cell methods are distinguished by being conservative: the fluid part of every cell is a well-defined control volume, and fluxes are accounted for on its actual boundary. Their classical difficulties are the description of the cut geometry, the small-cell problem, and, for moving bodies, the consistency between the change of the fluid volume of a cell and the flux swept by its wall, i.e. the discrete geometric conservation law.

The method described here addresses these points with a minimum of machinery. Its central object is the *closure identity* of a cut cell: the wet parts of its faces and its wall segment form a closed surface, so the wall area vector equals minus the sum of the wet face area vectors. Rather than computing the wall segment from the geometry and then enforcing consistency, the method defines the wall segment through the identity. Every consequence of the boundary condition — pressure force, no-penetration, wall traction, and for a moving body the volume swept — is then expressed with the same face fractions that the flux operators use, and the resulting discrete properties (exactness for uniform flow, no spurious force from a uniform pressure, exact compatibility of the pressure system, free-stream preservation) hold to round-off by construction.

The document is organised as follows. Section 2 states the equations, Section 3 defines the geometry and how it is computed, Section 4 derives the discretisation for a body at rest and its properties, and Section 5 the pressure–velocity coupling, the forces and the flow-rate forcing. Section 6 extends the method to moving bodies. Section 7 treats passive scalars. Section 8 derives the second-order wall treatment. Section 9 describes the implementation in OpenFOAM, Section 10 the verification, and Section 11 the limitations.

2 Governing equations

We solve the incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla p + \mathbf{g}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

with p the kinematic pressure, ν the kinematic viscosity and $\mathbf{g}$ a body force, in a domain Ω containing one or more rigid bodies B_i with prescribed velocity $\mathbf{u}_b(\mathbf{x}, t)$: zero for a body at rest, $\mathbf{u}_b = \mathbf{U}(t) + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{c}(t))$ for a rigid body with centre $\mathbf{c}(t)$. On each body surface one of two conditions is imposed: *no slip*, $\mathbf{u} = \mathbf{u}_b$, or *free slip*, $(\mathbf{u} - \mathbf{u}_b) \cdot \mathbf{n} = 0$ together with zero tangential traction, $(\mathbf{I} - \mathbf{n}\mathbf{n}) \cdot (\nabla \mathbf{u} + \nabla \mathbf{u}^\top) \cdot \mathbf{n} = \mathbf{0}$. Passive scalars T_k , $k = 1, \dots, K$, obey

$$\frac{\partial T}{\partial t} + \nabla \cdot (\mathbf{u}T) - \nabla \cdot (D\nabla T) = q, \quad (2)$$

each with its own constant diffusivity D , volumetric source q and, on each body, either a fixed value $T = T_w$ or zero diffusive flux $\partial T / \partial n = 0$. The coupling is one-way: the flow does not depend on the scalars.

The discretisation is a collocated finite-volume method with SIMPLE(C) pressure–velocity coupling in the arrangement of OpenFOAM’s `simpleFoam` (steady) and `pimpleFoam` (transient, implicit Euler with outer iterations). We write V_P for the volume of cell P , $\mathbf{S}_f$ for the area vector of face f pointing out of P , N for the neighbour across f , ϕ_f for the volume flux through f , $\mathbf{d}_f = \mathbf{x}_N - \mathbf{x}_P$ and $\delta_f = 1/|\mathbf{d}_f|$ for the inverse of the centre-to-centre distance. Face values $(\cdot)_f$ are linear interpolates, $\frac{1}{2}[(\cdot)_P + (\cdot)_N]$ on a uniform mesh. Sums $\sum_f$ run over the faces of the cell under consideration.

3 Geometric description of the body

3.1 Level set, fluid fractions and cell classes

Definition 1 (Level set and fractions). *A body is given by a function $\varphi(\mathbf{x})$, negative inside the body and positive in the fluid; only its sign and its zero level are used, and $\varphi = 0$ counts as solid. For a face f and a cell P ,*

$$\theta_f = \frac{|\{\mathbf{x} \in f : \varphi > 0\}|}{|\mathbf{S}_f|} \in [0, 1], \quad \alpha_P = \frac{|\{\mathbf{x} \in P : \varphi > 0\}|}{V_P} \in [0, 1] \quad (3)$$

are the fluid area and volume fractions. Several bodies are combined through the pointwise minimum of their level sets, which represents their union exactly.

Cells fall into three classes (Fig. 1a): *fluid* ($\alpha_P = 1$), *solid* ($\alpha_P = 0$) and *cut* ($0 < \alpha_P < 1$). A fourth set, overlapping the others, collects the *wall cells*: every cell with $\alpha_P > 0$ whose wall area (defined below) exceeds $10^{-10}V_P^{2/3}$. It contains all cut cells and, in addition, the fluid cells one of whose faces has been closed by a solid neighbour (Section 3.4). Wall cells are the only cells in which a wall term appears.

The fluid part Ω_P of a cut cell is a polyhedron bounded by the wet parts of the cell faces, of area vectors $\theta_f \mathbf{S}_f$, and by the wall segment, of area A_{wall} and outward unit normal $\mathbf{n}_w$ (Fig. 1b). Its volume is $\alpha_P V_P$.

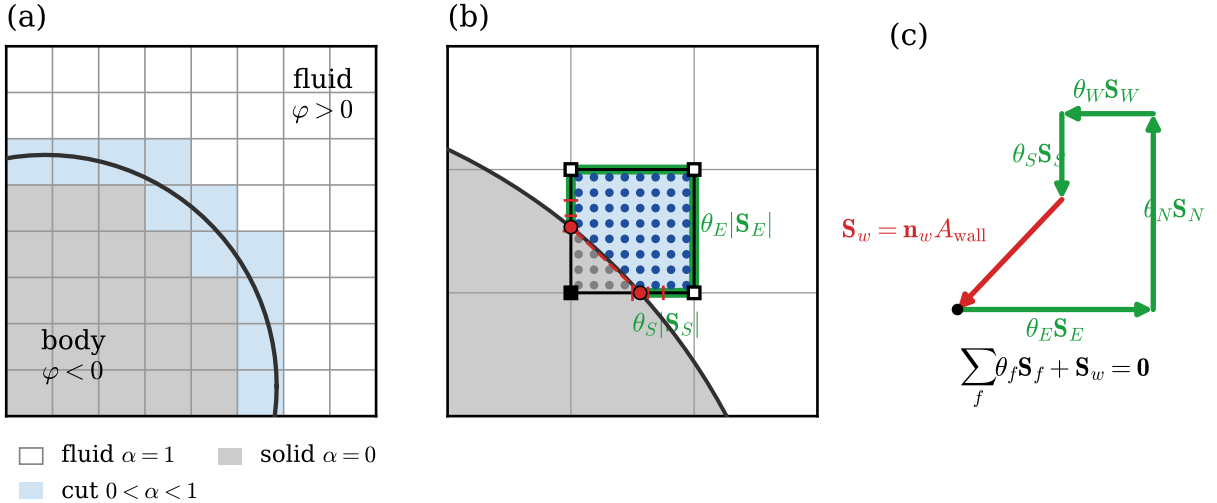


Figure 1: Geometry of the cut cells, computed with the algorithm of Section 3.3 for a cylinder of radius 1 on a mesh of spacing $h = 0.2$. (a) Cell classes. (b) One cut cell: corner signs (open squares fluid, filled squares solid), the bisection iterates (red ticks) converging to the edge crossings (red dots), the wet parts of the faces (green), the chord joining the crossings (dashed), and the sub-sampling lattice that gives α_P (blue points in the fluid, grey in the body; 8×8 drawn, 32×32 used). (c) The closure polygon of the same cell: the wet face area vectors, laid head to tail, are closed by the wall area vector $\mathbf{S}_w$ (Proposition 1).

3.2 The closure identity

Proposition 1 (Closure). *For every cell, $\sum_f \theta_f \mathbf{S}_f + \mathbf{n}_w A_{\text{wall}} = \mathbf{0}$.*

Proof. Apply the divergence theorem to a constant vector field $\mathbf{a}$ over Ω_P : $0 = \int_{\Omega_P} \nabla \cdot \mathbf{a} \, dV = \mathbf{a} \cdot (\sum_f \theta_f \mathbf{S}_f + \mathbf{n}_w A_{\text{wall}})$ for every $\mathbf{a}$. For a planar wall segment $\mathbf{n}_w A_{\text{wall}}$ is its area vector; for a curved one it is the integral $\int_{\text{wall}} \mathbf{n} \, dS$, and the identity holds unchanged. $\square$

We write

$$\mathbf{S}_w \equiv - \sum_f \theta_f \mathbf{S}_f = \mathbf{n}_w A_{\text{wall}}, \quad A_{\text{wall}} = |\mathbf{S}_w|, \quad (4)$$

for the wall area vector, pointing from the fluid into the body. In the implementation $\mathbf{S}_w$ is never computed from the level set: it is *defined* by (4) from the same θ_f that weight every flux operator, so the identity holds to round-off however the wall is inclined or curved and, for a moving body, however fast it moves. Figure 1c shows the construction. Two consequences are used throughout. First, every property that follows from the divergence theorem for a closed surface holds discretely: a constant field has zero net flux, a uniform pressure exerts no force. Second, the geometric errors of the fractions (the chord instead of the arc on a face, the quadrature error of α_P) change *where* the discrete wall is, but never make the discrete control volume leak.

In a cell whose faces are all fully open or fully closed, (4) also gives a non-zero $\mathbf{S}_w$ as soon as one face is closed: for a fluid cell next to a solid neighbour, $\mathbf{S}_w$ is the area vector of the closed face, i.e. the cell has a staircase wall. The two cases need no distinction in any operator.

3.3 Computation of the fractions

The fractions are computed once for bodies at rest and at every time step when a body moves, as follows.

1. *Evaluation.* The union level set is evaluated at the mesh points, the face centres and the cell centres.
2. *Edge crossings.* On every edge whose end points have different fluid status, the crossing is located by bisection of the exact level set, $n_{\text{bis}} = 40$ steps by default. The edge is always traversed in a canonical direction, from the lexicographically smaller end point A to the larger one B : with bracket $[\lambda_{\text{lo}}, \lambda_{\text{hi}}] \subset [0, 1]$, the midpoint $\mathbf{x}_m = A + \frac{1}{2}(\lambda_{\text{lo}} + \lambda_{\text{hi}})(B - A)$ replaces the end of the bracket that has the same fluid status, and the crossing is the final midpoint. Its error is at most $2^{-41} \approx 4.5 \times 10^{-13}$ of the edge length. The canonical direction makes the two copies of an edge on a processor boundary, which have identical coordinates, obtain bit-identical crossings; across a translational periodic boundary the lexicographic order is preserved, so the two sides agree whenever the level set is itself periodic. All cut edges are bisected together, one level-set call per iteration, which matters for bodies whose level set is expensive (triangulated surfaces).
3. *Face fractions.* A face whose vertices are all fluid has $\theta_f = 1$, one whose vertices are all solid $\theta_f = 0$. Otherwise the vertices are walked in order; each fluid vertex is appended to a polygon, and after every vertex whose status differs from the next the crossing of that edge is appended. The polygon area $|\sum_i \frac{1}{2}(\mathbf{p}_i - \mathbf{p}_0) \times (\mathbf{p}_{i+1} - \mathbf{p}_0)|$ divided by $|\mathbf{S}_f|$ is θ_f . A face with alternating signs (a saddle) is resolved as *fluid-connected*: the walk yields one polygon joining both fluid corners. The wet part of the face is thus approximated by the polygon whose edges are chords of the wall's trace on the face, a second-order approximation in the mesh spacing. The centroid $\mathbf{x}'_f$ of the polygon, by fan triangulation with the triangles weighted by their area projected on the face normal, and the fraction are kept as computed, before absorption and synchronisation (the geometric cut, θ_f^{geo}), for the rotating-wall source of Section 6.3 and the wall pieces of Section 8.5.
4. *Cell fractions.* A cell is a candidate for sub-sampling when the level set at its centre, its face centres and its vertices does not have one sign; the others have $\alpha_P = 1$ or 0. For a candidate, α_P is the fraction of fluid points in a lattice of n_s points per direction ($n_s = 32$ by default) placed at the sub-cell centres of the cell's bounding box,

$$\alpha_P = \frac{\#\{k : \varphi(\mathbf{x}_k) > 0\}}{\#\{k : \mathbf{x}_k \in P\}}, \quad (5)$$

where samples outside the cell are discarded (the point-in-cell test is skipped for axis-aligned hexahedra, whose bounding box is the cell). In a direction that is not solved (the empty direction of a two-dimensional case) a single sample at the cell centre is taken, so a two-dimensional case samples n_s^2 points and a three-dimensional one n_s^3 . This midpoint quadrature has an error of order $1/n_s$ per cell for a generic cut (about 10^{-3} for $n_s = 32$), which is small compared with the discretisation error but, as Section 6.2 shows, not small enough to be differentiated in time.

5. *Sliver absorption and coupled faces*, Section 3.4.

6. *Derived quantities*, Section 3.5.

Two features smaller than a cell are not resolved: a body that crosses a face without crossing any of its edges leaves θ_f unchanged, and one that enters a cell without reaching its centre, a face centre or a vertex leaves $\alpha_P = 1$. Note also that α_P and θ_f come from different algorithms and are therefore not exactly consistent (the polyhedron implied by the θ_f does not have exactly the volume $\alpha_P V_P$); nothing in the method requires them to be, because α_P enters only as a volume weight and $\mathbf{S}_w$ is built from θ_f alone.

3.4 Sliver absorption and coupled faces

A cut cell with a tiny fluid volume but open faces is the source of the small-cell problem. The fractions are therefore post-processed in two steps (Fig. 2).

Absorption. Every cell with $\alpha_P < \alpha_{\min}$ ($\alpha_{\min} = 0.02$ by default) is absorbed into the body: $\alpha_P := 0$ and *all its faces are closed*, $\theta_f := 0$. The same rule closes the faces of every cell that has $\alpha_P = 0$ from the start, so a solid cell never has an open face. Closing the faces is essential: an open face between an absorbed cell and a fluid neighbour would let the neighbour exchange flux with a cell that carries no equation, and the closure identity would then place the neighbour's wall on a face that is not a wall. After absorption the neighbours see the closed face through (4) as part of their wall (Fig. 2b), so the treatment is automatically consistent. The absorption displaces the boundary by at most $O(\alpha_{\min} h)$ and removes the fluid volume $\alpha_P V_P < \alpha_{\min} V_P$ of each absorbed cell. The pass is not iterated: a neighbour that loses faces keeps its α .

Coupled faces. The two copies of a face on a periodic or processor boundary must carry the same fraction. After absorption, θ_f is synchronised across coupled faces with the minimum of the two copies,

$$\theta_f := \min(\theta_f, \theta_{f'}), \quad (6)$$

which also propagates the closing of an absorbed cell across the boundary. The order matters (Fig. 2c): synchronising first and absorbing second would close the face on the side of the absorbed cell only, and its partner cell would keep an open face into a cell without equation. Section 10.2 shows the effect on a body crossing a periodic seam. A difference larger than 10^{-10} between the two copies before synchronisation is reported as a warning (a body that is not periodic although the mesh is).

3.5 Derived quantities

From the fractions the following fields are formed.

- The wall vector $\mathbf{S}_w$ by (4), accumulated face by face: each internal face subtracts $\theta_f \mathbf{S}_f$ from its owner and adds it to its neighbour, each boundary face subtracts it from its cell. Boundary faces of the domain count with their own θ_f , so a body that cuts an inlet or a wall patch still has a closed control volume. The front and back faces of a two-dimensional case do not enter (they cancel).
- The wall area $A_{\text{wall}} = |\mathbf{S}_w|$ and the wall normal $\mathbf{n}_w = \mathbf{S}_w / A_{\text{wall}}$.

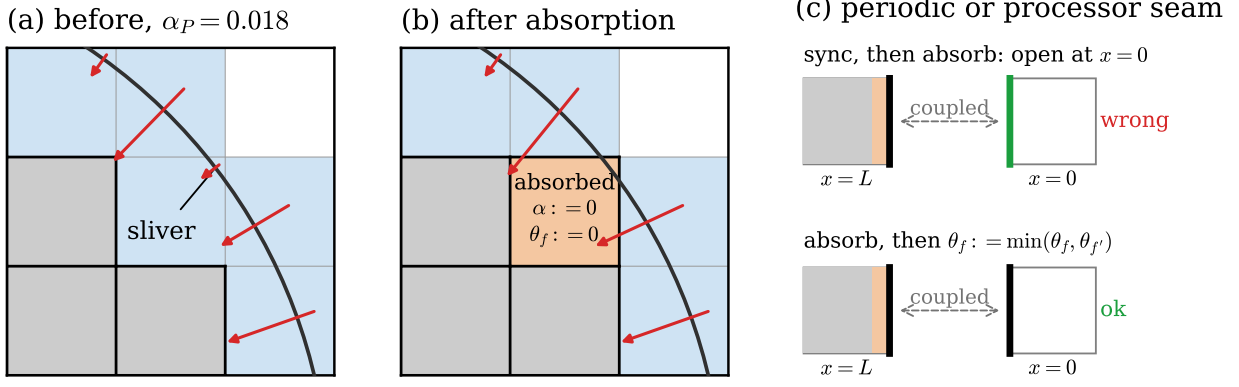


Figure 2: Sliver absorption, with real fractions for a cylinder of radius 1 on a mesh of spacing 0.2. Closed faces are black lines; red arrows are the wall vectors $\mathbf{S}_w$ of the wall cells (scaled by 0.8), drawn from the fluid centroid. (a) A sliver with $\alpha_P = 0.018$. (b) After absorption its faces are closed, and the neighbour above it becomes a wall cell whose $\mathbf{S}_w$ now includes the closed face. (c) On a periodic or processor boundary the coupled face must be closed on both sides, which requires absorbing *before* synchronising with the minimum.

- The *wall distance* of the cell value from the wall,

$$d_{\text{wall}} = \frac{\alpha_P V_P}{2A_{\text{wall}}}. \quad (7)$$

It is exact for a fluid slab: if the fluid part of a cell is a layer of thickness δ on a wall of area A parallel to a face, then $\alpha_P V_P = A\delta$, $A_{\text{wall}} = A$, and $d_{\text{wall}} = \delta/2$ is the distance from the centroid of the layer to the wall. For a general cut it is an estimate of that distance from two integral quantities that the closure identity delivers exactly; Fig. 3 compares it with the true centroid distance. For a fluid cell with one closed face on a Cartesian mesh it gives $h/2$, the exact distance of the cell centre from the face.

- The *no-slip (Dirichlet) coefficient*

$$c_{\text{ns}} = \frac{A_{\text{wall}}}{d_{\text{wall}} V_P} = \frac{2A_{\text{wall}}^2}{\alpha_P V_P^2} \quad [\text{m}^{-2}] \quad (8)$$

in wall cells of no-slip bodies (and of wall cells without a body), zero elsewhere. Multiplied by a diffusivity it is the wall exchange coefficient per unit volume.

- The *blanking coefficient* $b_P = \sum_f |\mathbf{S}_f| \delta_f / V_P$ in solid cells, zero elsewhere: the viscous diagonal per unit volume and diffusivity of an unobstructed cell.
- The body index of every cut, solid and wall cell (Section 3.6).

3.6 Bodies

Table 1 lists the body descriptions. Every body carries a wall condition (`wallType noSlip` or `freeSlip`, and per scalar a fixed value or zero gradient) and may be inverted ($\varphi \rightarrow -\varphi$, e.g. fluid inside a cylinder).

A closed triangulated surface (any OpenFOAM `searchableSurface` with an inside/outside test, e.g. `triSurfaceMesh` reading an STL file) uses the signed distance to the surface as level set, so that the same bisection applies. A surface without a volume type is rejected. A body given as a fluid-fraction field α is intended for fractions produced by another model (e.g. a deposit or a particle, re-read every step when the body is flagged as moving). Its cell fractions are the field values (clipped to $[0, 1]$, optionally converted from a solid fraction); point values are obtained by

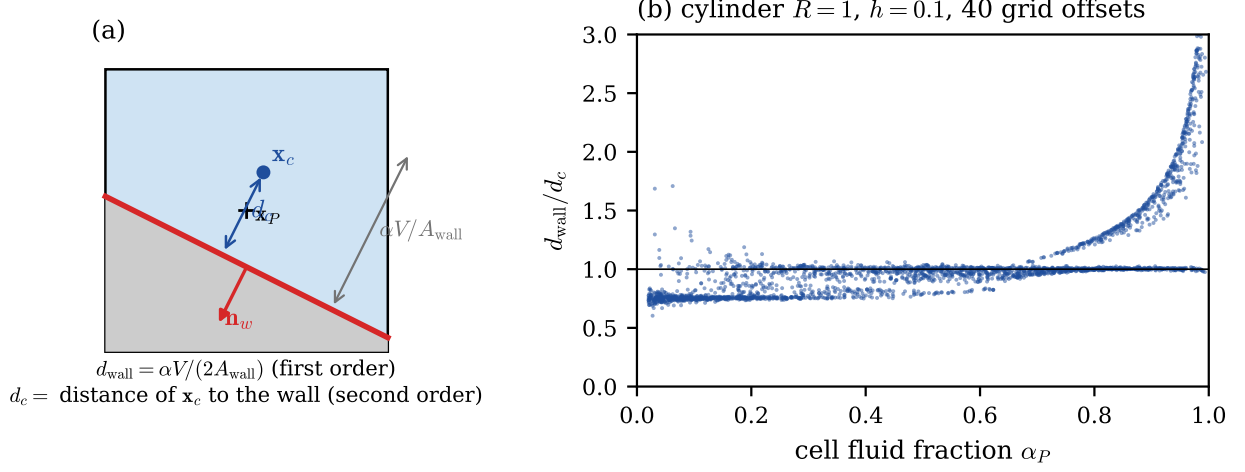


Figure 3: Wall distance. (a) In a cut cell the first-order estimate (7) is half the thickness $\alpha V / A_{\text{wall}}$ of the slab with the same volume on the same wall area; the distance d_c of the fluid centroid $\mathbf{x}_c$ from the wall is the quantity it approximates. The second-order treatment of Section 8 uses d_c itself. (b) Ratio of the two over all cut cells of a cylinder of radius 1 for 40 random offsets of a mesh of spacing 0.1: the estimate is within 25% for $\alpha_P \lesssim 0.8$ and overestimates the distance in cells cut near a corner, whose small wall area makes d_{wall} large (and c_{ns} small, so the wall term there is weak but also unimportant).

Table 1: Body descriptions. $\mathbf{c}$ centre, R radius, $\hat{\mathbf{a}}$ unit axis, $\mathbf{h}$ half-size and $\mathbf{q} = |\mathbf{x} - \mathbf{c}| - \mathbf{h}$ componentwise, $\hat{\mathbf{n}}$ unit normal into the fluid.

type	level set $\varphi(\mathbf{x})$	remark
cylinder	$ \mathbf{r} - (\mathbf{r} \cdot \hat{\mathbf{a}})\hat{\mathbf{a}} - R, \mathbf{r} = \mathbf{x} - \mathbf{c}$	infinite
sphere	$ \mathbf{x} - \mathbf{c} - R$	
box	$ \max(\mathbf{q}, \mathbf{0}) + \min(\max_i q_i, 0)$	exact signed distance
plane	$(\mathbf{x} - \mathbf{x}_0) \cdot \hat{\mathbf{n}}$	half space
any closed searchableSurface	$\pm \mathbf{x} - \mathbf{x}_{\text{nearest}} , -$ inside	e.g. STL
alphaField	— (fraction field supplied)	see text

inverse-distance interpolation, $\alpha - \frac{1}{2}$ serves as a point level set, edge crossings are found by linear interpolation of that level set, and the same polygon construction gives θ_f . This route smears the wall over about one cell (Section 11).

The level sets of all bodies with a level set are combined by their minimum before the fractions are computed, so overlapping bodies form an exact union; fraction-field bodies are merged afterwards with the minimum of α and of θ . Every cut or solid cell is attributed to the body with the most negative level set at its centre (for a fraction-field body: every cell whose fraction it lowered), and a wall cell that is not cut inherits the body across its closed face. The attribution selects the wall condition, the wall velocity, the body to which the cell's force contribution is added, and the scalar wall condition. A wall cell without a body is treated as no-slip and its force is added to the first body.

4 Discretisation for a body at rest

Every term of (1) is integrated over the fluid part Ω_P of each cell. The boundary of Ω_P consists of the wet parts of the faces and of the wall segment. The face parts are treated as in any finite-volume method, weighted by θ_f ; the wall segment contributes to each term according to the boundary condition, and this is the only place where the boundary condition enters. Table 2 collects the result, and the subsections derive each entry.

Table 2: Volume-integrated terms on the fluid part of a cell for a body at rest; Γ is a diffusivity, p_f and $\bar{\mathbf{u}}_f$ are linear face interpolates, ϕ_f is the wet face flux.

term	wall contribution	discrete operator
unsteady	—	$\alpha_P V_P (\mathbf{u}_P^{n+1} - \mathbf{u}_P^n) / \Delta t$
convective	zero ($\mathbf{u} \cdot \mathbf{n}_w = 0$)	$\sum_f \phi_f \bar{\mathbf{u}}_f, \quad \phi_f = \theta_f \bar{\mathbf{u}}_f \cdot \mathbf{S}_f$
viscous, free slip	zero (no tangential traction)	$\sum_f \theta_f \Gamma_f \delta_f \mathbf{S}_f (\mathbf{u}_N - \mathbf{u}_P)$
viscous, no slip	$\nu (\mathbf{0} - \mathbf{u}_P) A_{\text{wall}} / d_{\text{wall}}$	same, plus $-\nu c_{\text{ns}} V_P \mathbf{u}_P$
pressure	$p_P \mathbf{S}_w$, absorbed by closure	$\sum_f \theta_f (p_f - p_P) \mathbf{S}_f$
continuity	zero	$\sum_f \phi_f = 0$

4.1 Term-by-term derivation

Unsteady term. $\int_{\Omega_P} \partial_t \mathbf{u} dV = \alpha_P V_P d\mathbf{u}_P / dt$ for a cell whose fluid volume does not change, discretised with implicit Euler. The cell value $\mathbf{u}_P$ is the average over the fluid part.

Convective term. $\oint_{\partial\Omega_P} \mathbf{u} (\mathbf{u} \cdot \mathbf{n}) dS$ receives nothing from the wall, where $\mathbf{u} \cdot \mathbf{n}_w = 0$ for a wall at rest under either condition. On the wet face parts the mass flux is $\phi_f = \theta_f \bar{\mathbf{u}}_f \cdot \mathbf{S}_f$ and the transported value is the face interpolate (central differencing in all cases reported here, any OpenFOAM convection scheme in general).

Viscous term. $\oint \nu \nabla \mathbf{u} \cdot \mathbf{n} dS$ over the wet face parts is the standard two-point flux weighted by θ_f , with the diffusivity interpolated to the face and multiplied by θ_f ,

$$\sum_f \theta_f \nu_f |\mathbf{S}_f| \frac{\mathbf{u}_N - \mathbf{u}_P}{|\mathbf{d}_f|} \quad (+ \text{ the usual non-orthogonal correction on non-orthogonal meshes}). \quad (9)$$

On the wall segment, free slip means zero tangential traction and, since $\mathbf{u} \cdot \mathbf{n}_w = 0$, no normal viscous flux either, so the wall contributes nothing. For no slip the wall shear is approximated

by a two-point difference between the cell value and the wall value $\mathbf{0}$ over the distance (7),

$$\int_{\text{wall}} \nu \nabla \mathbf{u} \cdot \mathbf{n} \, dS \simeq -\nu A_{\text{wall}} \frac{\mathbf{u}_P - \mathbf{0}}{d_{\text{wall}}} = -\nu c_{\text{ns}} V_P \mathbf{u}_P. \quad (10)$$

This is one implicit diagonal term per no-slip wall cell and per velocity component; the components stay uncoupled because the difference is taken for the full vector (the normal component is small anyway, being driven to zero by continuity).

Pressure term. The volume-integrated gradient is $\oint_{\partial\Omega_P} p \mathbf{n} \, dS$; splitting the boundary into the wet faces and the wall segment, and taking $p_w = p_P$ on the wall (first order),

$$\nabla_V p|_P \equiv \oint p \mathbf{n} \, dS = \sum_f \theta_f p_f \mathbf{S}_f + p_P \mathbf{S}_w \stackrel{\text{Prop. 1}}{=} \sum_f \theta_f (p_f - p_P) \mathbf{S}_f. \quad (11)$$

The wall pressure force — the form drag — is therefore absorbed exactly into the face sum. The momentum equation uses $\nabla_V p/V_P$, i.e. the gradient per unit *cell* volume; the factor α_P belongs to the mobility, not to the gradient (Section 5.2).

Continuity. $\oint \mathbf{u} \cdot \mathbf{n} \, dS = \sum_f \phi_f + \int_{\text{wall}} \mathbf{u} \cdot \mathbf{n}_w \, dS$, and the wall integral vanishes for a wall at rest. Hence $\sum_f \phi_f = 0$ with the same wet fluxes as in the convective term.

4.2 Properties

The following properties hold to round-off because every wall-related quantity is built from the same θ_f .

Proposition 2 (No force from a uniform pressure). *For $p \equiv p_0$, $\nabla_V p = \mathbf{0}$ in every cell, cut or not.*

Proof. $p_f - p_P = 0$ in (11). □

Proposition 3 (Free-stream divergence). *For a uniform discrete velocity $\mathbf{u} \equiv \mathbf{U}$, $\sum_f \phi_f = -\mathbf{U} \cdot \mathbf{S}_w$ in every cell. In particular the discrete divergence vanishes in a cut cell if and only if $\mathbf{U} \cdot \mathbf{n}_w = 0$.*

Proof. $\sum_f \phi_f = \mathbf{U} \cdot \sum_f \theta_f \mathbf{S}_f = -\mathbf{U} \cdot \mathbf{S}_w$ by (4). □

Proposition 4 (No-penetration is implied, not imposed). *Let the velocity be locally uniform, $\mathbf{u} = \mathbf{U}$, on the faces of a wall cell with $A_{\text{wall}} \neq 0$. Discrete continuity $\sum_f \phi_f = 0$ holds if and only if $\mathbf{U} \cdot \mathbf{n}_w = 0$.*

Proof. Immediate from Proposition 3. The statement holds for any shape of the wall inside the cell, since only the area vector $\mathbf{S}_w$ enters. □

Proposition 4 is the structural point of the method: the no-penetration condition is never written down, it is a consequence of enforcing mass conservation on the cut cell with fluxes and wall segment taken from the same fractions. The free-slip condition appears only as the *absence* of a wall term in the viscous operator, and the no-slip condition as the presence of the single diagonal term $\nu c_{\text{ns}} V_P$.

Proposition 5 (Small cells strengthen the diagonal). *In a cut cell with a planar wall segment, the no-slip coefficient (8) grows like $1/\alpha_P$ as $\alpha_P \rightarrow 0$ at fixed A_{wall} .*

Proof. $c_{\text{ns}} = 2A_{\text{wall}}^2/(\alpha_P V_P^2)$. □

The small-cell limit therefore drives $\mathbf{u}_P$ onto the wall value and increases the diagonal dominance of the momentum rows instead of degrading them; together with sliver absorption it is what keeps the system well conditioned without cell merging. The price is paid by the pressure (Section 5.2, and Section 8.6).

Proposition 6 (Discrete conservation). *Summing the momentum equations of all cells, the convective and viscous face fluxes cancel in pairs, and the pressure term reduces to the pressure force on the bodies, $\sum_P \nabla_V p|_P = \sum_P p_P \mathbf{S}_{w,P} + (\text{domain boundary faces})$.*

Proof. Each internal face appears in the rows of its two cells with the same θ_f and opposite $\mathbf{S}_f$. In (11) write $\sum_f \theta_f p_f \mathbf{S}_f + p_P \mathbf{S}_w$; the face terms cancel pairwise. Solid cells have all faces closed and contribute nothing. $\square$

Consequently the force on the bodies is the reaction to exactly the wall terms assembled in the momentum equations (Section 5.5), and at a steady state in a periodic domain the total force equals the driving body force times the fluid volume. The same argument gives global mass conservation, $\sum_P \sum_f \phi_f = 0$, and it is what makes the pressure system compatible (Section 5.3).

Measured on manufactured solutions of Poisson and convection–diffusion problems on a disk, the free-slip (Neumann) treatment is second-order accurate and the no-slip (Dirichlet) treatment first-order accurate. Section 8 shows that the first-order error comes mainly from the diffusive fluxes through the faces of cut cells, which assume the two cell values to sit on the cell centres, and only secondarily from the wall distance (7).

5 Pressure–velocity coupling, forces and flow-rate forcing

5.1 Solid cells

Every face of a solid cell is closed, so its rows in the momentum and pressure matrices have no off-diagonal entries and are decoupled from the fluid. They receive regular diagonals so that the linear solvers see a well-posed system.

- *Momentum.* The blanking diagonal $\nu b_P V_P = \nu \sum_f |\mathbf{S}_f| \delta_f$, the viscous diagonal of an unobstructed cell. The right-hand side is zero in a solid cell (the gradient (11) vanishes, the forcing is weighted with $\alpha_P = 0$), and the blanking source $\nu b_P V_P \mathbf{u}_b(\mathbf{x}_P)$ is added, so $\mathbf{u}_P = \mathbf{u}_b$ exactly, independently of the size of b_P .
- *Pressure.* The diagonal $-\bar{a}$, with $\bar{a}$ the mean magnitude of the diagonal over the non-solid cells, and a zero source, so $p_P = 0$.

Neither value affects the fluid. The coefficient of the momentum row matters only for the rate at which an iterative solver converges; its size makes the solid rows neither dominate nor spoil the scaling of the matrix.

5.2 SIMPLEC with cut-cell operators

Write the assembled momentum equation of cell P as

$$a_P \mathbf{u}_P = \mathbf{H}_P(\mathbf{u}) - \nabla_V p|_P, \quad \mathbf{H}_P = \mathbf{b}_P - \sum_{nb} a_{nb} \mathbf{u}_{nb}, \quad (12)$$

with $a_{nb} \leq 0$ the off-diagonal coefficients (OpenFOAM’s sign convention), $\mathbf{b}_P$ the explicit sources (old time level, wall and blanking sources, forcing) and $\nabla_V p$ the volume-integrated cut-cell gradient (11). With under-relaxation λ_U the diagonal is a_P/λ_U and $\mathbf{b}_P$ holds the relaxation source. Define, per unit volume as OpenFOAM does,

$$A_P = \frac{a_P}{V_P}, \quad H1_P = -\frac{1}{V_P} \sum_{nb} a_{nb}, \quad r_A = \frac{1}{A_P} = \frac{V_P}{a_P}, \quad r_{At} = \frac{1}{A_P - H1_P} = \frac{V_P}{a_P + \sum_{nb} a_{nb}}. \quad (13)$$

r_A is the mobility of SIMPLE and r_{At} that of SIMPLEC, the response of the cell velocity to a pressure gradient when the neighbours respond in the same way. Note that the momentum equation keeps the full cell volume in A : the fluid volume enters only through the unsteady diagonal $\alpha_P V_P / \Delta t$ and the wall term. The predictor is $\mathbf{H}/a_P$, and the Rhie–Chow flux of the predictor is

$$\phi_{H,f} = \theta_f (r_A \mathbf{H}/V)_f \cdot \mathbf{S}_f + \theta_f (r_{At} - r_A)_f |\mathbf{S}_f| \partial_n p_f^{\text{old}} \left[+ (r_A)_f \mathcal{D}_f \right], \quad (14)$$

with ∂_n the face-normal gradient of the **snGrad** scheme and $\mathcal{D}_f$ the transient Rhie–Chow correction (below). The pressure equation is the discrete divergence of the corrected flux, with the Laplacian weighted by the same θ_f as the flux,

$$\sum_f \theta_f (r_{At})_f |\mathbf{S}_f| \partial_n p_f = \sum_f \phi_{H,f} + S_P, \quad \phi_f = \phi_{H,f} - \theta_f (r_{At})_f |\mathbf{S}_f| \partial_n p_f, \quad (15)$$

where S_P is the mass source of a moving wall (zero for a body at rest, Section 6), and the cell velocity is corrected with the cut-cell gradient, $\mathbf{u}_P = r_A \mathbf{H}_P / V_P - (r_{At})_P \nabla_V p|_P / V_P + ((r_{At})_P - (r_A)_P) \nabla_V p^{\text{old}}|_P / V_P$ (the last two terms combine into the familiar SIMPLEC update with $\mathbf{H}/a_P$ modified by $(r_A - r_{At}) \nabla p^{\text{old}}$). Laplacian and flux in (15) carry the same θ_f , so a closed face transports neither flux nor pressure information, and ϕ_f is discretely divergence-free (or equal to $-S_P$) after the solve.

Floors of the mobility. In a cut cell dominated by its wall term, a_P can be close to $\sum_{nb} |a_{nb}|$ and the SIMPLEC denominator $A - H1$ tiny or negative. It is floored:

- transient solver: $A_P - H1_P \geq \alpha_P / \Delta t$, the unsteady diagonal of the cell’s own fluid volume (zero in solid cells, where $A - H1 = A > 0$);
- steady solver: $A_P - H1_P \geq 10^{-3} A_P$, a pure safeguard.

r_A itself is not floored. The reference implementation of Section 10 instead floors a_P at $\alpha_P V_P / \Delta t$; the difference is invisible in the converged transient solutions but is one reason why the steady solver converges to a slightly different fixed point (the Rhie–Chow dissipation depends on the relaxation).

Transient Rhie–Chow correction. With **ddtCorr** active, the flux of the predictor carries the difference between the old flux and the interpolate of the old velocity,

$$\mathcal{D}_f = c_f \frac{\phi_f^n - \theta_f \bar{\mathbf{u}}_f^n \cdot \mathbf{S}_f}{\Delta t}, \quad c_f = 1 - \min\left(\frac{|\phi_f^n - \theta_f \bar{\mathbf{u}}_f^n \cdot \mathbf{S}_f|}{|\phi_f^n| + \epsilon}, 1\right), \quad (16)$$

the Euler form of OpenFOAM’s **ddtCorr** with θ -weighted interpolation. It is added to (14) with the coefficient $(r_A)_f$. For a moving body the old flux lives on the old wet faces and is first rescaled to the new ones, $\phi_f^n \rightarrow \phi_f^n \theta_f^{n+1} / \theta_f^n$ where $\theta_f^n > 0$, and replaced by $\theta_f^{n+1} \bar{\mathbf{u}}_f^n \cdot \mathbf{S}_f$ (no correction) on faces that have just opened. For a rigid free stream the rescaled old flux equals the interpolated one and the correction vanishes. Other time schemes fall back to the Euler form with a warning.

5.3 Compatibility and the pressure reference

The fluid region is a pure Neumann island for the pressure: no face of the fluid carries a pressure boundary condition in a periodic domain, and the solid rows are decoupled. The pressure Laplacian L restricted to the fluid cells is therefore singular with null vector $\mathbf{1}$ and $\mathbf{1}^\top L = \mathbf{0}$. The reference is imposed as in **fvMatrix::setReference**, by doubling one diagonal entry d_r (and adding $d_r p_{\text{ref}}$ to the source) rather than replacing a row.

Lemma 1 (Transparent reference). *Let $L' = L + d_r \mathbf{e}_r \mathbf{e}_r^\top$ and $L'p = \mathbf{b}$ (with $p_{\text{ref}} = 0$). Then $d_r p_r = \mathbf{1}^\top \mathbf{b}$. In particular, for a compatible right-hand side ($\mathbf{1}^\top \mathbf{b} = 0$) the reference is transparent, $p_r = 0$ and $Lp = \mathbf{b}$ exactly; any incompatibility is deposited at the reference cell as a point source.*

Proof. Left-multiply $L'p = \mathbf{b}$ by $\mathbf{1}^\top$ and use $\mathbf{1}^\top L = \mathbf{0}$. $\square$

The right-hand side of (15) is compatible because $\sum_P \sum_f \phi_{H,f}$ telescopes (each internal and coupled face appears twice with opposite sign) and $\sum_P S_P = 0$ (Section 6.2). In a domain with inflow and outflow boundaries the steady solver additionally balances the boundary fluxes (`adjustPhi`). The reference cell must be a fluid cell; this is checked at start-up. For moving bodies the user's reference cell is kept while it is fluid and otherwise replaced by the non-solid cell with the largest level set at its centre, i.e. the cell farthest from the bodies, so that a body cannot swallow it (without a level set, the first fluid cell). The pressure level then jumps; forces on closed bodies are unaffected because $\sum_P \mathbf{S}_{w,P} = \mathbf{0}$ over a closed body.

5.4 Algorithm

Algorithm 1 summarises one time step of the transient solver; the steady solver is the same loop without the unsteady term, with one pass per iteration and relaxation of $\mathbf{u}$ and p . The lines marked $\triangleright$ exist only when a body moves (Section 6.4).

Algorithm 1: One time step of the transient cut-cell solver (`ibmPimpleFoam`).

```

1 if a body moves then //  $\triangleright$  moving bodies only
2   move the bodies to  $t^{n+1}$  and rebuild  $\theta_f, \alpha_P, \mathbf{S}_w, A_{\text{wall}}, d_{\text{wall}}, c_{\text{ns}}$ , cell sets;
3    $\mathbf{u}_b$  in the body-owned cells;  $S_P = \mathbf{u}_b \cdot \mathbf{S}_w$ ; store  $\alpha_P^n = \alpha_P^{n+1} - \Delta t S_P / V_P$ ;
4   re-interpolate the flux,  $\phi_f := \theta_f^{n+1} \bar{\mathbf{u}}_f^n \cdot \mathbf{S}_f$ ; relocate the pressure reference if needed;
5 for outer iteration  $k = 1, \dots, n_{\text{outer}}$  do
6   assemble  $\alpha V \partial_t \mathbf{u} + \nabla \cdot (\phi \mathbf{u}) - \nabla \cdot (\theta \nu \nabla \mathbf{u}) + \nu c_{\text{ns}} V \mathbf{u} + \nu b V \mathbf{u} = \mathbf{s}$ ;
7    $\triangleright$  add the sources  $\nu(c_{\text{ns}} + b)V \mathbf{u}_b$ ;
8   solve the predictor with  $-\nabla_V p$  (optional);
9   for corrector  $j = 1, \dots, n_{\text{corr}}$  do
10     $r_A, r_{At}$  (floored),  $\mathbf{H}/a_P, \phi_H$  by (14) with  $\theta_f$ ;
11    solve  $\sum_f \theta_f(r_{At})_f |\mathbf{S}_f| \partial_n p_f = \sum_f \phi_{H,f} + S_P$  ( $\triangleright$ ) with regular solid rows and the reference;
12     $\phi := \phi_H - \theta r_{At} |\mathbf{S}_f| \partial_n p$ ; correct  $\mathbf{u}$  with  $\nabla_V p$ ; update the flow-rate forcing;
13 solve the passive scalars with the final  $\phi$  (Section 7);

```

5.5 Force on the body

By closure, the pressure force on a body is the sum of the wall terms of (11),

$$\mathbf{F}_p = \sum_{P \in \text{wall}} p_P \mathbf{S}_{w,P}, \quad \mathbf{F}_v = \sum_{P \in \text{wall}} \nu c_{\text{ns}} V_P (\mathbf{u}_P - \mathbf{u}_b), \quad (17)$$

the viscous force being the reaction to the traction (10) assembled in the momentum equation (zero for free slip). Wall cells without a body are counted on the first body. The forces are per unit density; for a two-dimensional case they refer to the depth of the mesh, and are divided by it to obtain forces per unit depth. By Proposition 6, at a steady state in a periodic domain with forcing $\mathbf{g}$ the total force equals $\mathbf{g} \sum_P \alpha_P V_P$ exactly; this identity is used as a consistency check in Section 10.

5.6 Constant flow rate in periodic domains

A periodic domain is driven by a uniform body force $\mathbf{g} = g \mathbf{e}$ adjusted so that the bulk velocity $\bar{u} = \sum_P \alpha_P V_P \mathbf{e} \cdot \mathbf{u}_P / V_f$, $V_f = \sum_P \alpha_P V_P$, equals a target U . The force acts on the fluid only: the source of cell P is $g \mathbf{e} \alpha_P V_P$.

Gain. Adding a force increment δg changes the cell velocity by $\delta \mathbf{u}_P = \mathbf{e} \rho_P \alpha_P \delta g$, where ρ_P is the mobility of the cell to a force that is the same in the neighbours. For a uniform increment the neighbour couplings cancel in the momentum equation, so $\rho_P = 1/(A_P - H1_P) = r_{At}$, the SIMPLEC mobility. The bulk response is

$$\delta \bar{u} = \bar{r} \delta g, \quad \bar{r} = \frac{\sum_P (r_{At})_P \alpha_P^2 V_P}{V_f}, \quad \text{hence} \quad \delta g = \lambda \frac{U - \bar{u}}{\bar{r}}, \quad (18)$$

with a relaxation λ (1 by default); the extra factor α_P in $\bar{r}$ comes from the force being weighted with α_P and the average with $\alpha_P V_P$. After every momentum predictor and pressure correction, δg is computed, added to the accumulated increment, and the velocity is corrected by $\mathbf{e} (r_{At})_P \alpha_P \delta g$. The stock OpenFOAM controller uses the gain $1/A$, which underestimates the bulk response by the diffusion number $\nu \Delta t / h^2$ when the viscous term is implicit, and overshoots by two orders of magnitude in the cases of Section 10. The mobility is floored at $A - H1 \geq \max(10^{-3} A, \alpha_P / \Delta t)$ in transient runs, the second bound covering the cells emerging from a moving body. The flow direction may be given separately, so a zero target (a body moving through fluid at rest on average) is allowed.

6 Moving bodies

6.1 What the wall sweeps

The mesh never moves: no face carries a mesh velocity and no mesh-motion term appears in any operator, so the method is not an arbitrary Lagrangian–Eulerian scheme. Only θ_f , α_P and $\mathbf{S}_w$ depend on time, and they are recomputed from the level set at every time step. The entire modification lives on the wall segment.

Let $\Omega_P(t)$ be the fluid part of cell P . Its boundary consists of the wet parts of the faces, which lie in fixed planes and sweep no volume, and the wall segment, which moves with $\mathbf{u}_b$. The Reynolds transport theorem for the volume of $\Omega_P(t)$ gives

$$\frac{d(\alpha_P V_P)}{dt} = \int_{\text{wall}} \mathbf{u}_b \cdot \mathbf{n}_w dS \simeq \mathbf{u}_b \cdot \mathbf{S}_w, \quad (19)$$

and continuity over the same region, with $(\mathbf{u} - \mathbf{u}_b) \cdot \mathbf{n}_w = 0$ on the wall under either condition,

$$0 = \oint_{\partial \Omega_P} \mathbf{u} \cdot \mathbf{n} dS = \sum_f \phi_f + \int_{\text{wall}} \mathbf{u}_b \cdot \mathbf{n}_w dS \implies \boxed{\sum_f \phi_f = - \int_{\text{wall}} \mathbf{u}_b \cdot \mathbf{n}_w dS \equiv -S_P}, \quad (20)$$

with $\phi_f = \theta_f \bar{\mathbf{u}}_f \cdot \mathbf{S}_f$ the wet face flux as before. For a translation $\mathbf{u}_b$ is uniform and $S_P = \mathbf{u}_b \cdot \mathbf{S}_w$ exactly; a rotation needs more care (Section 6.3). The sign is as it must be: a body advancing into the fluid has $\mathbf{u}_b \cdot \mathbf{n}_w < 0$ ($\mathbf{n}_w$ points into the body), the cell loses volume, $S_P < 0$, and the net outflow through its faces is positive.

Table 3 lists every term. The convective row is the one that is easy to get wrong: the absolute mass flux through the wall is not zero, but the momentum balance over the *time-varying* region is

$$\frac{d}{dt} \int_{\Omega_P(t)} \mathbf{u} dV + \oint_{\partial \Omega_P(t)} \mathbf{u} (\mathbf{u} - \mathbf{w}) \cdot \mathbf{n} dS = \dots, \quad (21)$$

Table 3: Every term for a body at rest and a body moving. Only three rows differ.

term	body at rest	body moving
continuity	$\sum_f \phi_f = 0$	$\sum_f \phi_f = -S_P, \quad S_P = \mathbf{U} \cdot \mathbf{S}_w + S_P^{\text{rot}}$
convective	no wall term	no wall term
pressure	$p_P \mathbf{S}_w$, absorbed by closure	same
viscous, free slip	no wall term	no wall term
viscous, no slip	$\nu(\mathbf{0} - \mathbf{u}_P)A_{\text{wall}}/d_{\text{wall}}$	$\nu(\mathbf{u}_b - \mathbf{u}_P)A_{\text{wall}}/d_{\text{wall}}$
unsteady	$\alpha_P V_P(\mathbf{u}^{n+1} - \mathbf{u}^n)/\Delta t$	$(\alpha_P^{n+1} \mathbf{u}^{n+1} - \alpha_P^n \mathbf{u}^n)V_P/\Delta t$

with $\mathbf{w}$ the velocity of the boundary: on the wall $\mathbf{w} = \mathbf{u}_b$ and $(\mathbf{u} - \mathbf{u}_b) \cdot \mathbf{n} = 0$, on the fixed faces $\mathbf{w} = \mathbf{0}$ and the flux is the absolute one. So the convective operator is untouched, provided the unsteady term is the two-level one of Table 3 with the volume inside the difference. For free slip nothing else is added: repeating the argument of Proposition 4 with (20) forces $\mathbf{U} \cdot \mathbf{n}_w = \mathbf{u}_b \cdot \mathbf{n}_w$ for a locally uniform field — *the mass source is the boundary condition*. For no slip the wall traction $\nu(\mathbf{u}_b - \mathbf{u}_P)A_{\text{wall}}/d_{\text{wall}}$ keeps its diagonal half unchanged and acquires the source $\nu_{c_{\text{ns}}} V_P \mathbf{u}_b$, which vanishes for a wall at rest.

6.2 Which quantity is primary

Equations (19) and (20) are the same statement, so on paper either may be taken as the definition. Discretely the two choices are not equivalent, and only one of them works.

Geometry first. Compute α^n and α^{n+1} from the level set and set $S_P = (\alpha_P^{n+1} - \alpha_P^n)V_P/\Delta t$. This differentiates a sampled quantity: the sub-sampled α of (5) carries a quadrature error of order 10^{-3} per cell, absorption removes up to $\alpha_{\min} V_P$ in single steps, and divided by Δt both are amplified without bound. The resulting source is not compatible ($\sum_P S_P \neq 0$), so by Lemma 1 the whole defect is deposited at the pressure reference cell. Figure 4 quantifies both defects for a cylinder translating on a mesh: the per-cell discrepancy between the two sources is of the order of the source itself, and the domain sum of the geometric source is 10^{-2} instead of zero. In the reference implementation this variant gives a free-stream error of 18% of the body velocity at $u_b = 0.37$ and 90% at $u_b = 2$, which does not decrease with the number of outer iterations: it is a wrong fixed point, not a splitting error.

Flux first. Compute $\mathbf{S}_w$ from θ^{n+1} , set

$$S_P = \mathbf{U} \cdot \mathbf{S}_w + S_P^{\text{rot}}, \quad \text{then} \quad \alpha_P^n V_P := \alpha_P^{n+1} V_P - \Delta t S_P, \quad (22)$$

with $\mathbf{U}$ the translational velocity of the body and S_P^{rot} the flux of its rotation through the wall faces, (24) below (zero for a translating body). Nothing is differentiated, and three properties follow.

1. *Consistency with the flux.* For a uniform $\mathbf{u} \equiv \mathbf{u}_b$ the discrete divergence equals $-S_P$ *identically* (Proposition 3), because $\mathbf{S}_w$ is assembled from the same θ_f that the flux uses.
2. *Compatibility.* For the translational part, with $\mathbf{U}$ extended to all cells (its value off the wall cells is irrelevant since $\mathbf{S}_w = \mathbf{0}$ there),

$$\sum_P \mathbf{U} \cdot \mathbf{S}_{w,P} = \sum_{f \text{ internal}} \theta_f \mathbf{S}_f \cdot (\mathbf{U}_N - \mathbf{U}_O) = 0, \quad (23)$$

with O and N the owner and neighbour of f : $\sum_P S_P$ is zero to round-off for every body position and every pattern of absorbed slivers (an absorbed cell has $\mathbf{S}_w = \mathbf{0}$, coupled faces carry the same θ_f on both sides). The rotational part is a sum of face fluxes and telescopes in the same way (Proposition 8). The sum is not exact when two bodies with different velocities share a cut cell or a face, or when a moving body is cut by a non-periodic domain boundary.

3. *Geometric conservation law.* The second half of (22) is a backward Euler integration of (19): the classical resolution of the discrete geometric conservation law, *take the volume from the flux, not the flux from the volume*, applied per step. Since α^n is recomputed from the current geometry at every step and never accumulated, it cannot drift; it differs from the geometric fraction at the old time by $O(\Delta t^2)$ per step, i.e. it is first-order consistent.

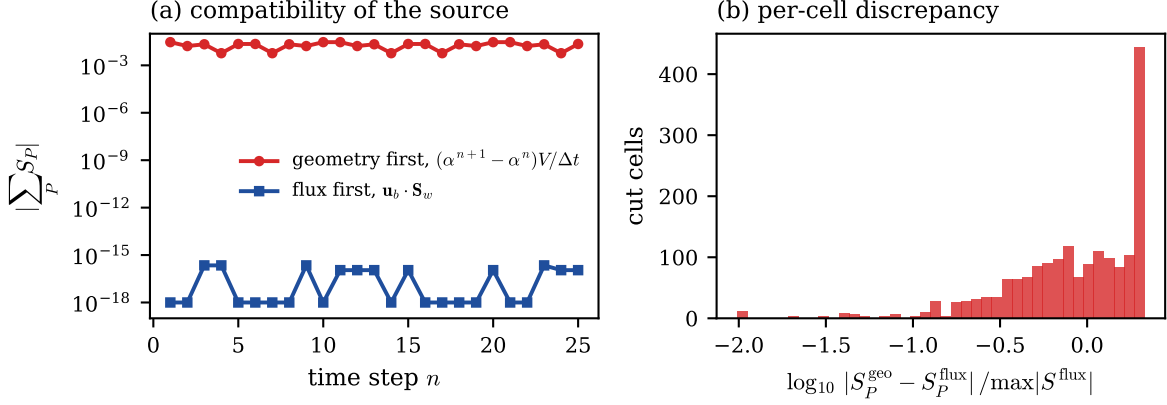


Figure 4: Geometry-first against flux-first mass source for a cylinder of radius 1 translating at $\mathbf{u}_b = (0.5, 0)$ on a mesh of spacing 0.1 with $\Delta t = 0.02$, computed with the fractions of Section 3.3. (a) Domain sum of the source per step: the flux form telescopes to round-off, the geometric form carries the sub-sampling error and the absorption jumps divided by Δt . (b) Distribution over the cut cells of the per-cell difference between the two, relative to the largest source.

Proposition 7 (Free-stream preservation). *Let the body translate at a constant velocity $\mathbf{u}_b$, let $\mathbf{u}^n \equiv \mathbf{u}_b$ in every cell (including the solid) and $p \equiv 0$, and let the flow-rate forcing be either absent or target $\mathbf{u}_b$. Then $\mathbf{u}^{n+1} \equiv \mathbf{u}_b$, $p \equiv 0$ solves the discrete system exactly, for any Δt , mesh and wall condition.*

Proof. In the momentum row of a fluid cell the contributions for $\mathbf{u} \equiv \mathbf{u}_b$ are: unsteady $(\alpha^{n+1}\mathbf{u}_b - \alpha^n\mathbf{u}_b)V_P/\Delta t$; convective $\mathbf{u}_b \sum_f \phi_f = -S_P\mathbf{u}_b$ (the flux having been formed from $\mathbf{u}_b$ through the new θ_f , Section 6.4); viscous zero, since the θ -weighted Laplacian annihilates a constant; no-slip wall $\nu c_{ns}V_P(\mathbf{u}_P - \mathbf{u}_b) = \mathbf{0}$; pressure zero by Proposition 2. The row reduces to $((\alpha^{n+1} - \alpha^n)V_P/\Delta t - S_P)\mathbf{u}_b = \mathbf{0}$, which is (22). In solid cells $\mathbf{u} = \mathbf{u}_b$ is the blanking solution. Continuity holds because $\sum_f \phi_f = -S_P$ identically, so the pressure equation has zero right-hand side and $p \equiv 0$; the transient Rhie–Chow correction vanishes because the rescaled old flux is the free-stream flux. $\square$

A rotating or accelerating body has no uniform exact solution, so the proposition does not apply to it; the construction still guarantees consistency (item 1) and compatibility (item 2).

Remark 1. With central differencing the diagonal contribution of the convective operator is $\frac{1}{2} \sum_f \phi_f = -\frac{1}{2}S_P$, so once continuity is satisfied the unsteady and convective terms together put $\frac{1}{2}(\alpha^{n+1} + \alpha^n)V_P/\Delta t$ on the diagonal, the *average* of old and new fluid volume, and their row sum is $\alpha_P^n V_P/\Delta t$. A cell just emerging from the body still carries half its new volume on the diagonal. In the first outer iteration of a step the re-interpolated flux does not yet satisfy (20) unless the velocity is uniform, so these statements hold from the second outer iteration on.

6.3 Where the body velocity is evaluated

For a translating body $\mathbf{u}_b$ is uniform, $S_P = \mathbf{U} \cdot \mathbf{S}_w$ is exact and the question does not arise. For a rotating one, $\mathbf{u}_b(\mathbf{x}) = \mathbf{U} + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{c})$ with $\mathbf{c}(t)$ the current centre of rotation, no single evaluation point gives the flux through the wall.

Lemma 2 (Rigid motion along the normal). *For any $\mathbf{x}$ and s , $\mathbf{u}_b(\mathbf{x} + s\mathbf{n}_w) \cdot \mathbf{n}_w = \mathbf{u}_b(\mathbf{x}) \cdot \mathbf{n}_w$.*

Proof. $(\boldsymbol{\omega} \times s\mathbf{n}_w) \cdot \mathbf{n}_w = 0$. □

Lemma 3 (Planar surfaces). *For a planar surface Γ with area vector $\mathbf{S}_\Gamma$ and centroid $\mathbf{x}_\Gamma$ and any field q linear in $\mathbf{x}$, $\int_\Gamma q \mathbf{n} dS = q(\mathbf{x}_\Gamma) \mathbf{S}_\Gamma$; in particular $\int_\Gamma \mathbf{u}_b \cdot \mathbf{n} dS = \mathbf{u}_b(\mathbf{x}_\Gamma) \cdot \mathbf{S}_\Gamma$.*

In two dimensions the wall of a cut cell is the chord between two edge crossings and $\mathbf{S}_w$ is normal to it (Fig. 1c). For a body of revolution turning about its axis the chord midpoint $\mathbf{x}_m$ lies on the line through $\mathbf{c}$ along $\mathbf{S}_w$, so the exact flux through the chord vanishes, while a point evaluation gives $\mathbf{u}_b(\mathbf{x}^e) \cdot \mathbf{S}_w = \boldsymbol{\omega} \cdot ((\mathbf{x}^e - \mathbf{x}_m) \times \mathbf{S}_w)$, which by Lemma 2 depends only on the offset of $\mathbf{x}^e$ across the normal. The cell centre, the centre moved along the normal and the fluid centroid all lie off that line in general: each leaves spurious local sources of order $|\boldsymbol{\omega}|hA_{\text{wall}}$, alternating in sign around the body, which checkerboard the cut-cell pressure. (An earlier version of the code evaluated $\mathbf{u}_b$ at such a point; for a cylinder turning in place the source was non-zero in every wall cell.)

The source is therefore split into the translational part, which is kept, and the flux of the rotation computed from the faces:

$$S_P = \mathbf{U} \cdot \mathbf{S}_w + S_P^{\text{rot}}, \quad S_P^{\text{rot}} = - \sum_{f \in P, \theta_f > 0} \theta_f (\boldsymbol{\omega} \times (\mathbf{x}'_f - \mathbf{c})) \cdot \mathbf{S}_f, \quad (24)$$

with $\mathbf{S}_f$ pointing out of P , θ_f the fraction after post-processing, and $\mathbf{x}'_f$ the centroid of the geometric wet part of the face (the fan-triangulated polygon of Section 3.3, before absorption); in periodic directions $\mathbf{x}'_f - \mathbf{c}$ is the minimum image. Because $\boldsymbol{\omega} \times (\mathbf{x} - \mathbf{c})$ is linear and divergence-free, the divergence theorem over the fluid part of the cell and Lemma 3 give $S_P^{\text{rot}} = \int_{\text{wall}} \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{c}) \cdot \mathbf{n} dS$ exactly, where the wall is that of the post-processed cell: the chord(s) of the geometric cut and the geometric wet parts of the faces closed by sliver absorption (Section 3.4). The translational part sees the same wall, since $\mathbf{S}_w$ is formed with the post-processed θ_f .

Proposition 8 (Rotating-wall source). *The source (24)*

1. *reduces to $\mathbf{U} \cdot \mathbf{S}_w$ for $\boldsymbol{\omega} = \mathbf{0}$, so free-stream preservation (Proposition 7) and every translating case are unchanged;*
2. *is compatible for every rigid motion and every mesh, $\sum_P S_P = 0$: every open face enters its two cells with opposite $\mathbf{S}_f$ and the same θ_f and $\mathbf{x}'_f$, and the two copies of a periodic face map to the same minimum image about $\mathbf{c}$;*
3. *vanishes, for a body of revolution turning about its axis, in every wall cell not next to a cell removed by the post-processing; next to one it is the flux of the rotation through the closed face, which moves with the body like the rest of the wall.*

Item 3 is a compromise. The purely geometric alternative (all geometrically wet faces, with the flux of every removed cell handed to its fluid neighbours in proportion to the shared wet area) vanishes in every cell of a body turning in place, but it leaves in a cell next to a removed one the rotation flux through the closed face, $O(|\boldsymbol{\omega}|R\theta_f^{\text{geo}}h)$, which that cell's continuity, with no flux through a closed face, cannot carry; an area-weighted share of the removed cell's flux does not cancel it face by face. The pressure takes up the mismatch in cells of small mobility: in the Taylor–Couette case of Section 10.5 the cut-cell pressure errors at the rotating wall grew three- to sixfold, and with the second-order treatment the velocity lost its second order. With (24) the closed face is part of the moving wall, as in the translational part, and the continuity of a cell is consistent with a velocity that moves with the body near the wall. For a cylinder turning in place on an 81^2 mesh, 5 of 63 wall cells carry a source.

The body velocity at a point is still needed for the *tangential* wall velocity of the no-slip traction $\nu(\mathbf{u}_b - \mathbf{u}_P)A_{\text{wall}}/d_{\text{wall}}$, the viscous force and the blanking. Solid cells use their centre and wall cells the point

$$\mathbf{x}_P^e = \mathbf{x}_{c,P} + d_{\text{wall}} \mathbf{n}_w, \quad (25)$$

the foot of the normal from the fluid centroid $\mathbf{x}_{c,P}$ (Section 8.1), which is what the two-point wall gradient differences against; the centroid is computed whether or not the second-order treatment is switched on. For a nearly full cell with a very small wall segment, $d_{\text{wall}} = \alpha_P V_P / (2A_{\text{wall}})$ is an effective distance that can exceed the cell size by orders of magnitude; taken literally, the point would lie far outside the cell, in a periodic domain possibly beyond the minimum image. Since the wall of such a cell lies inside it, the distance along $\mathbf{n}_w$ is limited to where the normal leaves the bounding box of the cell; the traction coefficient $A_{\text{wall}}/(d_{\text{wall}} V_P)$ keeps d_{wall} .

6.4 Algorithm

One time step (Algorithm 1 and Fig. 5):

1. Advance the time and move the bodies to t^{n+1} (implicit in time); keep θ_f^n for the transient Rhie–Chow correction.
2. Rebuild all geometry from scratch: θ_f , α_P (with sliver absorption and coupled-face synchronisation), $\mathbf{S}_w$, A_{wall} , d_{wall} , c_{ns} , the cell sets and the body attribution, the centroids and, if switched on, the second-order quantities. The old geometry is not kept.
3. Evaluate $\mathbf{u}_b$ in every body-owned cell (wall cells at (25), solid cells at their centre), form $S_P = \mathbf{u}_b \cdot \mathbf{S}_w$ and $\alpha_P^n V_P = \alpha_P^{n+1} V_P - \Delta t S_P$, and store α^n as the old-time value of α .
4. *Re-interpolate the carried flux*: $\phi_f := \theta_f^{n+1} \bar{\mathbf{u}}_f^n \cdot \mathbf{S}_f$. The flux returned by the previous step lives on the faces of the *old* cut mesh: a face that has just opened has no history there, and one that has just closed carries a history that no longer exists. The re-interpolated flux is only the initial guess of the convective operator — the flux the step returns is the Rhie–Chow projected one either way — but for a fluid translating rigidly with the body it lands exactly on the fixed point, which is why Proposition 7 holds from the first outer iteration. Without it the fixed point is approached at roughly a factor of two per outer iteration. The previous Rhie–Chow flux is kept as the old-time flux for (16).
5. Keep the pressure reference cell if it is fluid, otherwise relocate it (Section 5.3).
6. For each of n_{outer} outer iterations: assemble and solve the momentum predictor with the sources $\alpha^n V_P \mathbf{u}^n / \Delta t$, $\nu c_{\text{ns}} V_P \mathbf{u}_b$ (no slip) and the blanking source $\nu b_P V_P \mathbf{u}_b$; form $\mathbf{H}/a_P$ and ϕ_H ; solve (15) with S_P on the right-hand side; correct flux and velocity. The geometry is frozen during the outer iterations.

Equation (15) with S_P is the entire algorithmic modification of the pressure equation. Newly uncovered cells enter the step with the velocity they had in the solid, $\mathbf{u}_b(\mathbf{x}_P)$, and $p = 0$. For this reason velocity and pressure are *not* set to zero in the solid at start-up when a body moves: a free stream must start with $\mathbf{u} = \mathbf{u}_b$ in the solid.

6.5 Step restriction

The row sum of the unsteady plus convective operator is $\alpha_P^n V_P / \Delta t$, which becomes negative when the swept fraction does. By (22), $\alpha_P^n < 0$ when

$$\Delta t \mathbf{u}_b \cdot \mathbf{n}_w > \frac{\alpha_P^{n+1} V_P}{A_{\text{wall}}} = 2d_{\text{wall}}, \quad (26)$$

i.e. at a *receding* wall (the rear of a body), in cells being uncovered whose linearised swept volume exceeds their new fluid volume. This is a condition on the smallest wet cells, hence set by α_{min}

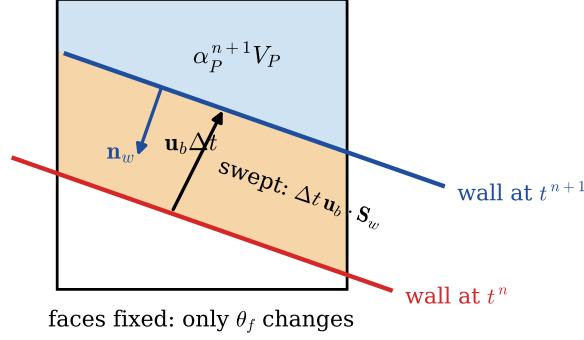


Figure 5: A cut cell over one step. The faces are fixed in space; only their wet fractions change. The shaded band is the volume the wall sweeps, $\Delta t \mathbf{u}_b \cdot \mathbf{S}_w = \Delta t S_P$ (negative here: the body advances and the cell loses volume), the only quantity the motion contributes to the discretisation; the old fluid volume is recovered from it, $\alpha_P^n V_P = \alpha_P^{n+1} V_P - \Delta t S_P$.

rather than by the mesh spacing. It is not fatal when violated in a few cells — the diagonal there is still dominated by the viscous and wall terms, which grow like $1/\alpha_P$ — and the solver reports $\min_P \alpha_P^n$ and the number of cells with $\alpha_P^n < 0$ at every step (in the oscillating-cylinder case of Section 10.2, one or two cells in 25 of 400 steps, $\min_P \alpha_P^n = -0.032$, Fig. 9). At an advancing staircase wall α_P^n can exceed 1 by the same mechanism; this is harmless.

6.6 Rigid body kinematics

A body is defined at rest and moved by a rigid motion: a translation $\mathbf{d}(t)$ with velocity $\mathbf{U}(t)$, and optionally a rotation with constant angular velocity $\boldsymbol{\omega}$ about the body’s reference centre $\mathbf{c}_0$ (the centre of an analytical body, the bounding-box centre of a triangulated one, or the point given as `centreOfRotation` in the motion dictionary), by the angle $|\boldsymbol{\omega}|t$ measured from $t = 0$. Table 4 lists the motions. The velocity must be the exact derivative of the displacement: the first supplies the wall condition, the second, through the geometry, the mass source, and if they disagree the two halves of the boundary condition pull against each other (this is not checked for prescribed functions).

Table 4: Body motions; every type accepts in addition a constant `angularVelocity` and a `centreOfRotation`.

type	displacement $\mathbf{d}(t)$	velocity $\mathbf{U}(t)$
none	$\mathbf{0}$	$\mathbf{0}$ (moving if $\boldsymbol{\omega} \neq \mathbf{0}$)
translating	$\mathbf{U}t$	$\mathbf{U}$
oscillating	$\mathbf{A} \sin(2\pi ft + \varphi_0)$	$2\pi f \mathbf{A} \cos(2\pi ft + \varphi_0)$
prescribed	Function1 $\mathbf{D}(t)$	Function1 $\mathbf{V}(t)$
external	set from code every step	set from code every step

Level-set queries transform the query point into the body frame. With $\mathbf{c}(t) = \mathbf{c}_0 + \mathbf{d}(t)$ and periodic lengths L_k (given in the geometry dictionary),

$$\mathbf{x}_{\text{body}} = \mathbf{c}_0 + \mathbf{R}(-|\boldsymbol{\omega}|t) \left[\text{mi}(\mathbf{x} - \mathbf{c}(t)) \right], \quad \text{mi}(r)_k = r_k - L_k \left[r_k / L_k + \frac{1}{2} \right], \quad (27)$$

the minimum image of the point about the current centre in every periodic direction, so that a body may cross a periodic boundary. The same transform gives $\mathbf{u}_b(\mathbf{x}) = \mathbf{U} + \boldsymbol{\omega} \times \text{mi}(\mathbf{x} - \mathbf{c}(t))$. The `external` motion is the hook for a body driven by its own equation of motion; a body given as a fraction field and flagged as moving has its field re-read every step and its velocity taken from a supplied velocity field or a uniform value.

7 Passive scalar transport

7.1 Discrete equation

Integrating (2) over the fluid part of a cell with the same arguments as for momentum gives, for each scalar T ,

$$\begin{aligned} \frac{(\alpha_P^{n+1} T_P^{n+1} - \alpha_P^n T_P^n) V_P}{\Delta t} + \sum_f \phi_f \bar{T}_f - \sum_f \theta_f D \delta_f |\mathbf{S}_f| (T_N - T_P) \\ + D c_{w,P} V_P (T_P - T_{w,P}) + D b_P V_P (T_P - T_{w,P}) = q_P \alpha_P^{n+1} V_P + s_P V_P, \end{aligned} \quad (28)$$

where every T without superscript is at t^{n+1} and

- ϕ_f is the final wet flux of the flow solver, so the scalar is convected by exactly the flux that satisfies discrete continuity ($\sum_f \phi_f = -S_P$ for a moving body); the convection scheme is that of the case (central in all cases reported here);
- the diffusive face flux is θ_f -weighted as for momentum, with the constant diffusivity D of the scalar (plus the non-orthogonal correction of the Laplacian scheme);
- $c_{w,P} = A_{\text{wall}}/(d_{\text{wall}} V_P) = 2A_{\text{wall}}^2/(\alpha_P V_P^2)$ is the wall coefficient (8), applied in the wall cells of the bodies that are fixed-value *for this scalar*, with $T_{w,P}$ the value of the attributed body; it is zero in the wall cells of zero-flux bodies;
- b_P is the blanking coefficient of the solid cells, where $T_{w,P}$ is the value of the owning body if it is fixed-value for this scalar and 0 otherwise;
- q_P is an optional source per unit fluid volume, uniform in a disk (two dimensions) or sphere (three dimensions) of given centre and radius, set in the cells whose centre lies inside it; it multiplies α_P^{n+1} and never injects into the solid;
- s_P collects the sources and sinks of the finite-volume options (`fvOptions`) that list the scalar, per unit *cell* volume as OpenFOAM applies them; an implicit coefficient enters the diagonal, an explicit one the source;
- α_P^n is the swept fraction (22) for a moving body and $\alpha_P^n = \alpha_P^{n+1}$ for a body at rest.

The wall segment is seen either as a fixed-value surface — the analogue of no slip, one diagonal term $D c_w V_P$ and the source $D c_w V_P T_w$, i.e. the flux $D A_{\text{wall}} (T_P - T_w)/d_{\text{wall}}$ through the wall — or as a zero-flux surface, the analogue of free slip, which needs no term at all: the wall segment is not a mesh face and simply carries no flux. The two masks, for velocity and scalars, are independent, and each scalar has its own. Cells containing the wall are fluid cells: the scalar enters their fluid part through the wet fractions of their faces and never crosses the wall segment. There is no convective wall term, because the relative flux through the wall vanishes; in particular no mass-source term appears for a moving body, the effect of the moving wall being carried entirely by the two-level unsteady term. The solid rows are decoupled (all faces closed, $\alpha^{n+1} = \alpha^n = 0$) and reduce to $D b_P V_P (T_P - T_{w,P}) = s_P V_P$, so $T_P = T_{w,P}$ exactly when no option acts there; a cell released by a moving body starts from this defined value.

In OpenFOAM the equation is assembled as

```
fvm::ddt(ibm.alpha(), T) + fvm::div(phi, T) - fvm::laplacian(DT*ibm.theta(), T)
+ fvm::Sp(DT*wallCoeff, T) + fvm::Sp(DT*ibm.blankCoeff(), T)
== DT*(wallCoeff + ibm.blankCoeff())*Twall + q*ibm.alpha() + fvOptions(T)
```

with `wallCoeff` and `Twall` the per-scalar masks. The transient solver solves the scalars once per time step, after the last outer iteration, with the final flux; the steady solver once per SIMPLE iteration, after the pressure correction.

7.2 Discrete conservation

Proposition 9 (Scalar budget). *In a periodic or closed domain, with no finite-volume option acting in solid cells and the linear systems solved exactly, the scalar content $C^n = \sum_P \alpha_{g,P}^n V_P T_P^n$, with α_g the geometric fraction of the current step, satisfies*

$$C^{n+1} - C^n = \Delta t \sum_P (q_P \alpha_P^{n+1} + s_P) V_P - \Delta t \sum_P D_{c_w,P} V_P (T_P^{n+1} - T_{w,P}) - \sum_P (\alpha_{g,P}^n - \alpha_P^n) V_P T_P^n. \quad (29)$$

Proof. Sum (28) over all cells. Every internal and coupled face appears in the rows of its two cells with the same flux and the same diffusive coefficient and opposite sign, whatever the interpolation of $\bar{T}_f$, so the convective and diffusive sums vanish; the wall is not a face. The solid rows balance the blanking term against the options acting there, which are zero by assumption. What remains is $\sum_P (\alpha_P^{n+1} T_P^{n+1} - \alpha_P^n T_P^n) V_P / \Delta t = \sum_P (q \alpha^{n+1} + s) V_P - \sum_P D_{c_w} V_P (T - T_w)$, and $\sum_P \alpha_P^n V_P T_P^n = C^n - \sum_P (\alpha_g^n - \alpha^n) V_P T_P^n$. $\square$

Summed over the steps, (29) is the budget

$$\underbrace{C^n - C^0}_{\text{content}} - \underbrace{\sum_m \Delta t \sum_P (q_P \alpha_P + s_P) V_P}_{\text{injected}} + \underbrace{\sum_m \Delta t \sum_P D_{c_w,P} V_P (T_P - T_{w,P})}_{\text{through walls}} + \underbrace{\sum_m \sum_P (\alpha_P^{\text{prev}} - \alpha_P^m) V_P T_P^m}_{\text{swept defect}} = \text{round-off}, \quad (30)$$

which the solver evaluates and reports at every step, with the options' contribution computed exactly as the matrix applied it (implicit part times the new value, explicit part as assembled). The residual is the accumulated linear-solver tolerance. The steady solver reports the balance of rates instead, $\sum_P (q \alpha + s) V_P$ against $\sum_P D_{c_w} V_P (T - T_w)$. The swept-defect term exists only for moving bodies: α^n is integrated from the wall flux and is first-order consistent with the geometric fraction α^{prev} of the previous step, and a cell absorbed into the body hands over no content. It is the price of the exact geometric conservation law of Section 6.2; it is of either sign, of order Δt , and small (Section 10.4). Fluxes through non-periodic domain boundaries are not included in the reported budget; all cases reported here are periodic.

7.3 Consistency and boundedness

Conservation and consistency need different conditions. A uniform field $T \equiv T_0$ with $q = s = 0$ is preserved by (28) when every wall term vanishes, i.e. when the fixed-value walls have $T_w = T_0$: the convective row sum is $T_0 \sum_f \phi_f$, which cancels against the unsteady term by (22) and (20) for a moving body (Proposition 7), to the tolerance of the pressure solver. For a moving zero-flux body a released cell comes from the solid with the blanked value 0, so a uniform $T_0 \neq 0$ is perturbed there by $O(\alpha_P^n T_0)$, the same effect as the swept defect.

With central differencing the off-diagonal coefficient of face f is $\frac{1}{2} \phi_f - \theta_f D \delta_f |\mathbf{S}_f|$. The factor θ_f appears in both terms, so the condition for non-positive off-diagonals, the face Péclet number $|\bar{\mathbf{u}}_f \cdot \mathbf{S}_f| / (D \delta_f |\mathbf{S}_f|) < 2$, is the same in cut cells as in full ones. The diagonal is $\alpha^{n+1} V / \Delta t$ ($\frac{1}{2}(\alpha^{n+1} + \alpha^n) V / \Delta t$ for a moving body, plus the neighbour sum) and the wall coefficient, which grows like $1/\alpha_P$ in small cells (Proposition 5). The matrix is then an M-matrix and a discrete maximum principle, $\min(T^n, T_w) \leq T^{n+1} \leq \max(T^n, T_w)$ for $q = s = 0$, holds provided $\alpha^n \geq 0$, i.e. the step restriction of Section 6 is met. In the cases below the cell Péclet number is about 0.6. A bounded convection scheme (upwind or TVD) keeps conservation and consistency, since both rely only on the flux. In a steady problem with zero-flux walls only and a net source there is no steady solution; with fixed-value walls the level of the solution is set by the wall sink.

7.4 Several scalars and input

The scalars are listed in `constant/scalarTransportProperties` and solved in turn, each with its own diffusivity, source, wall conditions, options and budget:

```

scalars
{
  T { DT 0.1; source { centre (5 8.5 0); radius 0.5; rate 1; }
      walls { cylinder { type fixedValue; value 0; } ".*" { type zeroGradient; } } }
  c { DT 0.01; }           // walls from the body entries of ibmProperties
}

```

The body names in `walls` may be regular expressions; a body not listed falls back to its own `scalarWall` (default `zeroGradient`) and `scalarWallValue` entries in `constant/ibmProperties`. An older single-scalar form (the entries of one scalar at the top level of the file) is still accepted. The coupling is one-way: the flow does not see the scalars, and the scalars do not see each other except through options that read other fields, which are then lagged by the sequential solution.

8 Second-order wall treatment

The treatment of Sections 4–5 is first order for no-slip and fixed-value walls. This section derives an optional treatment, switched on by `secondOrder true` in `constant/ibmProperties`, that makes the diffusive operators and the velocity second order. It rests on one principle — the value of a cut cell lives at the centroid of its fluid part — and on two ingredients: a diffusive face flux consistent with the centroids, and the wall distance of the centroid. Neither needs a ghost value, a probe into the fluid or a larger stencil.

8.1 Cell values live at fluid centroids

For a smooth field, the average over the fluid part of a cell equals the point value at its centroid to second order,

$$\frac{1}{\alpha_P V_P} \int_{\Omega_P} u \, dV = u(\mathbf{x}_{c,P}) + O(h^2), \quad \mathbf{x}_{c,P} = \frac{1}{\alpha_P V_P} \int_{\Omega_P} \mathbf{x} \, dV, \quad (31)$$

because the linear term of the Taylor expansion about $\mathbf{x}_c$ integrates to zero. In a full cell $\mathbf{x}_c = \mathbf{x}_P$ and nothing changes; in a cut cell the centroid is displaced by $O(h)$ from the cell centre, and every operator that treats u_P as the value at $\mathbf{x}_P$ makes an $O(h)$ error in position. Two centroids are computed (Fig. 6):

- the *fluid centroid* $\mathbf{x}_{c,P} = \mathbf{x}_P + \mathbf{o}_P$ as the mean position of the fluid samples of the sub-sampling that gives α_P (5), with error $O(h/n_s)$; it is computed for every candidate cell (and set to $\mathbf{x}_P$ in full and solid cells), whether or not the switch is on;
- the *wet-face centroid* $\mathbf{x}'_f = \mathbf{x}_f + \mathbf{o}'_f$ as the centroid of the fluid polygon of Section 3.3, by fan triangulation, $\mathbf{x}'_f = \sum_i |\mathbf{a}_i|(\mathbf{p}_0 + \mathbf{p}_i + \mathbf{p}_{i+1}) / (3 \sum_i |\mathbf{a}_i|)$, exact for the polygon.

Fraction-field bodies have no sub-samples and keep $\mathbf{o}_P = \mathbf{0}$.

8.2 Why the no-slip wall is first order

Consider the diffusive flux density through face f between P and N , discretised as $(u_N - u_P)\delta_f$ with $\delta_f = 1/h$ on a uniform Cartesian mesh (variant F0). The values live at the centroids, separated by

$$\mathbf{d} = \mathbf{x}_{c,N} - \mathbf{x}_{c,P} = h \mathbf{n} + (\mathbf{o}_N - \mathbf{o}_P). \quad (32)$$

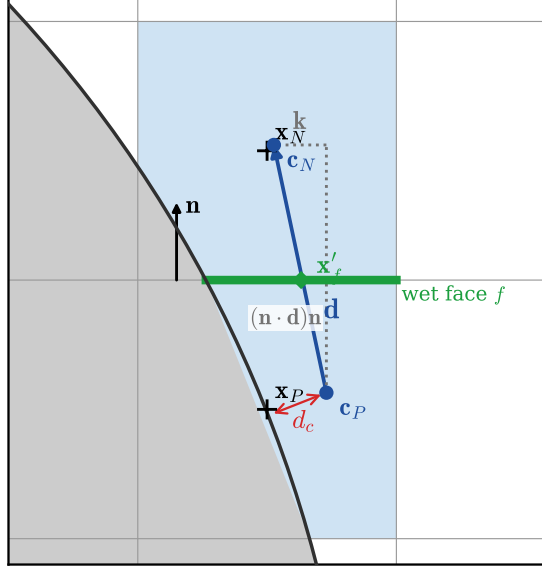


Figure 6: Second-order ingredients for two neighbouring cut cells P and N of a cylinder (real fractions, $h = 0.2$): cell centres $\mathbf{x}_P, \mathbf{x}_N$ (crosses), fluid centroids $\mathbf{c}_P = \mathbf{x}_{c,P}, \mathbf{c}_N$ (dots), the wet part of the common face with its centroid $\mathbf{x}'_f$, the centroid vector $\mathbf{d}$ with its normal part $(\mathbf{n} \cdot \mathbf{d})\mathbf{n}$ and the remainder $\mathbf{k}$ along the face, and the wall distance d_c of the centroid.

Expanding about the midpoint $\mathbf{m}$ of the two centroids, $u_N - u_P = \nabla u(\mathbf{m}) \cdot \mathbf{d} + O(|\mathbf{d}|^3)$, so

$$\frac{u_N - u_P}{h} = \mathbf{n} \cdot \nabla u(\mathbf{m}) + \frac{(\mathbf{o}_N - \mathbf{o}_P) \cdot \nabla u(\mathbf{m})}{h} + O(h^2). \quad (33)$$

The second term is an $O(1)$ error in the flux density on every face of the layer of cut cells, since the offsets are $O(h)$. Its size depends on the wall condition. The offsets of cut cells are predominantly normal to the wall: at a Neumann wall the normal derivative vanishes, so $(\mathbf{o}_N - \mathbf{o}_P) \cdot \nabla u = O(h) \cdot O(h)$ and the error is $O(h)$ per unit flux density; at a Dirichlet wall the normal derivative is $O(1)$ and the error is $O(1)$. This is the observation of the order study summarised in Fig. 14: the Neumann problems are second order with the original scheme, the Dirichlet problems first order, and improving only the wall distance (V1/F0) leaves them first order. The wall distance (7) is itself inconsistent for inclined walls (for the corner triangle $(0,0), (h,0), (0,h)$ it gives $0.177h$ against the true centroid distance $0.236h$, and Fig. 3b shows the spread), but that is not the limiting error.

8.3 Centroid-consistent diffusive flux

Decompose the face normal as

$$\mathbf{n} = \frac{\mathbf{d}}{\mathbf{n} \cdot \mathbf{d}} + \mathbf{k}, \quad \mathbf{k} = \mathbf{n} - \frac{\mathbf{d}}{\mathbf{n} \cdot \mathbf{d}}, \quad \mathbf{k} \cdot \mathbf{n} = 0, \quad (34)$$

so that $\mathbf{k}$ lies in the plane of the face (Fig. 6). Then $\mathbf{n} \cdot \nabla u = \mathbf{d} \cdot \nabla u / (\mathbf{n} \cdot \mathbf{d}) + \mathbf{k} \cdot \nabla u$, and the first part is approximated by the two-point difference along $\mathbf{d}$, which is centred between the two values, while the second is evaluated with a reconstructed gradient $\mathbf{g}_f$:

$$\mathbf{n} \cdot \nabla u|_f \simeq c_f (u_N - u_P) + [\mathbf{g}_f \cdot \mathbf{n} - c_f \mathbf{g}_f \cdot \mathbf{d}], \quad c_f = \frac{1}{\max(\mathbf{n} \cdot \mathbf{d}, 0.1 h_f)}, \quad (35)$$

with h_f the mesh distance across the face. This is the over-relaxed non-orthogonal correction of finite-volume methods, applied to the centroids instead of the cell centres. The flux through the wet part of the face is

$$F_f = \theta_f \Gamma_f |\mathbf{S}_f| \left\{ c_f (u_N - u_P) + [\mathbf{g}_f \cdot \mathbf{n} - c_f \mathbf{g}_f \cdot \mathbf{d}] \right\}, \quad (36)$$

with the first term implicit and the bracket explicit (deferred correction, re-evaluated at every assembly, i.e. once per outer iteration). The ingredients are:

- $\mathbf{d} = \mathbf{d}_f + \mathbf{o}_N - \mathbf{o}_P$ on internal and coupled faces that are open and between two non-solid cells; on all other faces the mesh values (which makes the correction vanish on faces between full cells of an orthogonal mesh);
- the *least-squares gradient on the centroids*,

$$\mathbf{g}_P = \mathbf{A}_P^{-1} \sum_N w_{PN} \mathbf{d}_{PN} (u_N - u_P), \quad \mathbf{A}_P = \sum_N w_{PN} \mathbf{d}_{PN} \mathbf{d}_{PN}^\top, \quad w_{PN} = |\mathbf{d}_{PN}|^{-2}, \quad (37)$$

over the neighbours across open faces (the wall value is not used), with a unit diagonal added for each non-solved direction; $\mathbf{g}_P = \mathbf{0}$ if $\mathbf{A}_P$ is singular (e.g. all open neighbours collinear);

- the face gradient $\mathbf{g}_f = \frac{1}{2}(\mathbf{g}_P + \mathbf{g}_N)$.

Proposition 10 (Exactness for linear fields). *If $u = a + \mathbf{G} \cdot \mathbf{x}$ and $\mathbf{A}_P, \mathbf{A}_N$ are non-singular, then (35) returns $\mathbf{n} \cdot \mathbf{G}$ exactly, independently of the floor on $\mathbf{n} \cdot \mathbf{d}$.*

Proof. The least-squares right-hand side is $\sum_N w \mathbf{d}(\mathbf{d} \cdot \mathbf{G}) = \mathbf{A}_P \mathbf{G}$, so $\mathbf{g}_P = \mathbf{g}_N = \mathbf{g}_f = \mathbf{G}$. Then $c_f \mathbf{G} \cdot \mathbf{d} + \mathbf{G} \cdot \mathbf{n} - c_f \mathbf{G} \cdot \mathbf{d} = \mathbf{G} \cdot \mathbf{n}$; the same c_f multiplies the implicit and the subtracted explicit term, so the floor cancels. $\square$

Further properties follow directly.

- *Conservation.* $\mathbf{n}$ and $\mathbf{d}$ change sign with the orientation of the face, so the flux (36) is antisymmetric and the discrete conservation of Proposition 6 is kept (also across coupled faces).
- *Uniform and linear fields* are reproduced, so the free-stream property of a body at rest and Proposition 7 are unaffected, and so are closure, blanking, the mass source and the flow-rate forcing.
- *Stability.* The implicit part has coefficients $\theta_f \Gamma_f |\mathbf{S}_f| c_f > 0$ and is an M-matrix; the floor keeps $c_f \leq 10/h_f$. The explicit remainder vanishes where $\mathbf{d} \parallel \mathbf{n}$.
- *Accuracy.* $\mathbf{g}_f = \nabla u(\mathbf{m}) + O(h)$ and $|\mathbf{k}| = O(1)$ in cut cells, so the flux density error on cut faces is $O(h)$ instead of the $O(1)$ of (33); on the $O(h^{1-d})$ faces of the cut layer this is the classical situation in which a local truncation error of first order in a codimension-one set still yields a second-order global error. The order study confirms it (Section 10.5).

In OpenFOAM, (35) is one `snGradScheme`, `cutCellCorrected`: its delta coefficients are c_f and its explicit correction the bracket. Selected for `laplacianSchemes` and `snGradSchemes`, it is used consistently by the Gauss Laplacian of the momentum and pressure equations (the correction moves to the source), by `pEqn.flux()`, and by `fvc::snGrad` in the SIMPLEC term of (14), so the flux and velocity corrections stay consistent without any change to the solvers. The scalar equations inherit it through their Laplacian scheme.

8.4 Wall flux with the centroid distance

In the no-slip term (10) (and the fixed-value scalar term) the distance d_{wall} of (7) is replaced by the distance of the fluid centroid from the wall,

$$d_c = \max(\varphi(\mathbf{x}_{c,P}), 10^{-3} V_P^{1/3}), \quad (38)$$

evaluated with the union level set (V1). It is the exact centroid distance for level sets that are signed distances (the analytical bodies and triangulated surfaces); fraction-field bodies keep (7).

For a fluid cell behind a closed face it is the distance to the true surface rather than to the closed face. The two-point wall flux

$$\Gamma A_{\text{wall}} \frac{u_P - u_w}{d_c} = \Gamma A_{\text{wall}} \left(\partial_n u|_w + \frac{1}{2} d_c \partial_n^2 u|_w + \dots \right) \quad (39)$$

has an $O(h)$ error in flux density on an area $O(h^{d-1})$, the same class as (35), and suffices for second order. A three-point normal probe (V3) gives no improvement, and scaling the wall row with the cell's Laplacian (V2) is inconsistent, because the cell Laplacian contains the tangential curvature (Fig. 14). The same d_c enters the viscous force and the evaluation point (25) of the wall velocity.

8.5 Pressure gradient and forces

With the switch on, the cut-cell gradient (11) is made exact for a linear pressure. Let $\mathbf{g}$ be the least-squares gradient (37) of p on the centroids, $\mathbf{g}_f = w_f \mathbf{g}_O + (1 - w_f) \mathbf{g}_N$ its face interpolate, and $\mathbf{m}_f = w_f \mathbf{x}_{c,O} + (1 - w_f) \mathbf{x}_{c,N}$ the point the linear interpolate $\bar{p}_f = w_f p_O + (1 - w_f) p_N$ of centroid values belongs to. Then

$$\nabla_V p|_P = \sum_f \theta_f \left[\bar{p}_f + \mathbf{g}_f \cdot (\mathbf{x}'_f - \mathbf{m}_f) \right] \mathbf{S}_f + \sum_k \left[p_P + \mathbf{g}_P \cdot (\mathbf{x}_k - \mathbf{x}_{c,P}) \right] \mathbf{S}_k. \quad (40)$$

The face correction moves the face value from $\mathbf{m}_f$ to the wet-face centroid $\mathbf{x}'_f$; it is applied on every open face of a cell with a centroid offset and on every partially wet face, and vanishes between two full cells of an orthogonal mesh ($\mathbf{m}_f = \mathbf{x}'_f = \mathbf{x}_f$). The wall of the cell is split into planar pieces k with area vector $\mathbf{S}_k$ (out of the fluid) and centroid $\mathbf{x}_k$:

- the wall polygon(s) of the geometric cut. On every cut face of the cell the wall trace runs from the crossing where the face boundary leaves the fluid to the next crossing; oriented by the face (reversed for the owner, whose $\mathbf{S}_f$ points out of the cell), the traces chain into closed loops, which are fan-triangulated about their mean point. In two dimensions a loop is the chord times the depth; in three dimensions it is the polygon of the cut-edge crossings, not planar in general, and the triangles are summed individually;
- the geometric wet part of every face closed by the post-processing, area vector $(\theta_f^{\text{geo}} - \theta_f) \mathbf{S}_f$ at the wet-face centroid.

By the closure of the geometric fluid polyhedron $\sum_k \mathbf{S}_k = \mathbf{S}_w$, which the code checks (the mismatch is 10^{-11} or less; a cell whose traces do not close would fall back on the extrapolation to the normal foot of the centroid, which has not occurred). Only $\sum_k (\mathbf{x}_k - \mathbf{x}_{c,P}) \mathbf{S}_k$ is stored, one tensor per wall cell, and the wall term is $p_P \mathbf{S}_w + \mathbf{g}_P \cdot \sum_k (\mathbf{x}_k - \mathbf{x}_{c,P}) \mathbf{S}_k$.

Proposition 11 (Linearly exact gradient). *If the least-squares gradient is exact for linear fields, (40) gives $\nabla_V p|_P = V_P^{\text{geo}} \mathbf{G}$ for every $p = p_0 + \mathbf{G} \cdot \mathbf{x}$ and every cell, with V_P^{geo} the volume of the geometric fluid polyhedron. It vanishes for a uniform p , conserves momentum, and the force $\mathbf{F}_P = \sum_P \sum_k p_k \mathbf{S}_k$ keeps the momentum-sum identity.*

Proof. Each face term is $p(\mathbf{x}'_f) \theta_f \mathbf{S}_f$, the exact integral over the planar wet face (Lemma 3); each wall term is $p(\mathbf{x}_k) \mathbf{S}_k$, exact over a planar piece. The wet faces and the pieces bound the geometric polyhedron, and the divergence theorem gives $V_P^{\text{geo}} \mathbf{G}$. A uniform p gives $p(\sum_f \theta_f \mathbf{S}_f + \sum_k \mathbf{S}_k) = \mathbf{0}$; the face corrections are antisymmetric because $\mathbf{x}'_f$ and $\mathbf{m}_f$ are shared by both cells. $\square$

The previous form corrected only partially wet faces, by $\mathbf{g}_f \cdot (\mathbf{x}'_f - \mathbf{x}_f)$, and took the wall value at the normal foot of the centroid, $p_c + d \mathbf{g} \cdot \mathbf{n}_w$; for a linear pressure its relative error in the wall cells was of order one (L_2 0.74–0.98, Section 10.5). The least-squares fit needs as many fluid neighbours as dimensions; a cell with fewer (a sliver in three dimensions) takes the mean

gradient of its fluid face neighbours, which keeps Proposition 11. The force (17) uses the same pieces. Since $\alpha_P V_P$ and V_P^{geo} differ by the sub-sampling error of α (Section 3.3), a fluid at rest under a uniform body force $\alpha V \mathbf{g}$ is in balance up to that error only; weighting the body force with V^{geo} , or computing α from the polyhedron, would make it exact.

8.6 The small-cell problem of the pressure

The operators and the velocity become second order with the two ingredients above, but the pressure of the cut cells does not converge (Section 10.5). The reason is the scaling of the mobility. In a thin cut cell the momentum diagonal is dominated by the wall term,

$$a_P \simeq \nu A_{\text{wall}}/d_c \propto \nu A_{\text{wall}}^2/(\alpha_P V_P), \quad r_{A,P} = V_P/a_P \propto \alpha_P V_P^2/(\nu A_{\text{wall}}^2), \quad (41)$$

against $r_A \sim h^2/\nu$ in a full cell, while the pressure row of the cut cell has the diagonal $\sum_f \theta_f(r_{At})_f |\mathbf{S}_f| c_f$, which scales with the wet fractions. The cut-cell pressure is thus determined by small flux residuals divided by a small coefficient, and the cell's own velocity barely responds to it, being pinned by the wall term. Through the face values p_f the error feeds back into the neighbours' momentum and pollutes the near-wall pressure, and with it the forces, at first order. The velocity is hardly affected. The consistency of the gradient is not the cause: with the linearly exact gradient of Section 8.5 the Taylor–Couette pressure errors change by at most 7% (Section 10.5). A cut-cell pressure that is not determined by the cut cell alone — cell merging or linking of cut cells below $\alpha \approx 0.5$ to a full neighbour, or a wall pressure formed from the full cells — is the natural next step.

8.7 Implementation and scope

- Switch and schemes: `secondOrder true` in `constant/ibmProperties`, and

```
laplacianSchemes { default Gauss linear cutCellCorrected; }
snGradSchemes   { default cutCellCorrected; }
```

Mismatched settings are not detected: `cutCellCorrected` with the switch off is the `uncorrected` scheme (identical to `corrected` on orthogonal meshes), and the switch with `corrected` gives `V1/F0`, which is still first order.

- With the switch off, θ , α , d_{wall} , the gradient, the forces and the schemes are bit-identical to the first-order treatment. The changes on this branch that are not behind the switch concern rotating bodies only: the evaluation point (25) of the wall velocity, which uses the centroid in both cases, and the rotational source (24).
- The convective face value is not reconstructed at the wet-face centroid: in the order study this refinement (C1) changes the error by 10–30% on coarse meshes and nothing asymptotically, and convection uses the scheme of the case.
- For moving bodies all centroid quantities and wall pieces are recomputed with the geometry at every step. The combination has been exercised only with a body rotating in place (Section 10.5).
- The scalars obtain (35) and (38) through the Laplacian scheme and the wall coefficient; the scalar tutorials still use the first-order schemes.
- Cost: one least-squares gradient per corrected field and component per assembly, the centroids, one coefficient and one offset per face, and one tensor per wall cell. The utility `ibmOrderTest` solves the manufactured problems of Section 10.5 with the library operators.

9 Implementation in OpenFOAM

9.1 Explicit operators on an unmodified mesh

The mesh is left untouched; θ_f and α_P enter the standard operators explicitly. The momentum equation of the transient solver reads

```
fvm::ddt(ibm.alpha(), U) // alpha V/dt; alpha.oldTime() = alpha^n
+ fvm::div(phi, U) // phi = theta (U_f . Sf) = ibm.flux(U)
- fvm::laplacian(ibm.thetaInterpolate(nu), U) // theta-weighted, no wall term
+ fvm::Sp(nu*ibm.noSlipCoeff(), U) // wall traction, no-slip bodies
+ fvm::Sp(nu*ibm.blankCoeff(), U) // blanking of solid cells
== fvOptions(U) - ibm.grad(p) // sum_f theta_f (p_f - p_P) Sf / V
```

and, for moving bodies, `UEqn -= ibm.movingWallSource(nu)` adds $\nu(c_{ns} + b)u_b$. The pressure equation is

```
fvm::laplacian(ibm.thetaInterpolate(rAtU), p) == fvc::div(phiHbyA) // phiHbyA = ibm.flux(HbyA)
pEqn -= ibm.massSource(); // moving bodies only: div(phi) = -S
ibm.constrainPressure(pEqn); // regular rows for the solid cells
```

and `pEqn.flux()` is then automatically θ -weighted. Everything else is `simpleFoam` and `pimpleFoam`: the boundary constraints on $\mathbf{H}/a_P$ and p , the pressure reference, pressure relaxation, finite-volume options, residual control and the PIMPLE loops. The steady and transient solvers differ from their stock counterparts only in the lines above, the floors of Section 5.2 and the forces; the moving-body operations sit behind `if (ibm.moving())` guards in the transient solver (geometry update and flux re-interpolation, moving-wall source, mass source, continuity report), and the steady solver rejects moving bodies.

This choice keeps the full cell volume V_P in `fvMatrix::A()`, so that $r_A = V_P/a_P$ is exactly the mobility of (13). Scaling the face area vectors and cell volumes of the mesh in place would make all operators θ/α -aware implicitly, but would change the SIMPLEC mobility to $\alpha V_P/a_P$, break the interpolation weights on closed faces and require floors for solid cells; explicit operators are transparent and term-for-term identical to the formulation above. The two-level unsteady term of Table 3 costs no solver code: the density-weighted time derivative `fvm::ddt(rho, U)` of OpenFOAM puts $\rho^{n+1}V/\Delta t$ on the diagonal and uses the old-time value of the weight in its source, so the swept fraction α^n of (22) is simply stored as the old-time value of α after the geometry update.

9.2 Structure

`ibmBody` a body (Table 1): analytical, any closed `searchableSurface`, or a supplied fraction field, with its wall condition and its `bodyMotion` (Table 4).

`cutCellGeometry` the fractions and derived fields of Section 3, the cut-cell operators (gradient, flux, θ -interpolation, `ddtCorr`), the solid-row regularisation, the forces, and `update()`, which returns immediately unless a body moves. The moving-only fields (body velocity, mass source, old face fractions) are allocated only then, so the static path carries no additional cost. It also holds the centroid geometry, the least-squares gradient and the wall pieces of Section 8.

`cutCellCorrected` the `snGradScheme` of (35).

`ibmMeanVelocityForce` the constant-flow-rate forcing of Section 5.6, as a finite-volume option.

`ibmScalarTransport` equation (28) for one scalar: the per-body wall masks, the disk source, the options listing the field, and the budget (30). `ibmScalarTransportList` holds the scalars of `scalarTransportProperties` and solves them in turn.

`ibmSimpleFoam`, `ibmPimpleFoam` the solvers.

`ibmSetGeometry` computes and writes the geometry fields, and checks that the cut-cell gradient of a uniform field is zero (the closure identity).

`ibmOrderTest` the manufactured problems of Section 10.5 on the library operators.

The geometry computation is local except for the synchronisation of coupled faces and global reductions, so parallel runs need nothing else.

9.3 Input and output

The geometry is controlled by `constant/ibmProperties`:

```
alphaMin    0.02;    // sliver threshold
nSubSamples 32;      // n_s, samples per direction for alpha
nBisect     40;      // bisection steps per cut edge
writeFields true;    writeForces true;
periodicLengths (10 10 0); // minimum image for bodies crossing periodic boundaries
secondOrder false;    // Section 7
bodies
{
    cylinder
    {
        type cylinder; centre (5 5 0); axis (0 0 1); radius 1;
        wallType noSlip; // or freeSlip; invert true for fluid inside
        motion { type oscillating; amplitude (1 0 0); frequency 0.05; }
        scalarWall fixedValue; scalarWallValue 0; // default for all scalars
    }
}
```

The flow-rate forcing is an `fvOptions` entry of type `ibmMeanVelocityForce` with the target `Ubar`, an optional `flowDir` and `relaxation`. The fields `ibmAlpha`, `ibmTheta`, `ibmSw`, `ibmWallDistance`, `ibmNoSlipCoeff` and `ibmBodyIndex` are written at start-up; for moving bodies α , θ , S_w , the body index and `ibmUb` are written at every write time, and `ibmCentroidOffset` when the second-order treatment is on. The forces of Section 5.5 are printed every step and written to `postProcessing/ibmForces/<time>/forces.dat` (time, then the pressure and viscous force of each body).

10 Verification

Unless stated otherwise the cases use a circular cylinder of radius $R = 1$ centred in a 10×10 doubly periodic box discretised by 81×81 cells ($h = 0.1235$), $\nu = 1$, $\alpha_{\min} = 0.02$, $n_s = 32$ and central differencing for convection. The reference is an independent implementation of the same discretisation on a structured grid (direct sparse solves, exact pinning of the bulk velocity by superposition), so that the comparison isolates the OpenFOAM implementation from the formulation. It differs from the OpenFOAM implementation in details that do not change the discrete fixed point: it uses the level set $r^2 - R^2$, a penalty of 10^{12} instead of the blanking diagonal, floors a_P instead of $A - H1$, and pins the bulk velocity exactly instead of by the controller of Section 5.6. Forces are quoted per unit depth and density.

10.1 Body at rest

The flow is driven at bulk velocity 0.5 ($Re_D = 1$). Table 5 compares geometry, fields and drag. The fractions agree to 10^{-11} and the cell fractions exactly. The transient solver run with the reference’s pseudo-time step ($\Delta t = 65h^2/\nu$, Euler, one corrector, SIMPLEC, no `ddtCorr`) converges to the same discrete fixed point (velocity to 10^{-5} , drag to 2×10^{-6}); the steady solver differs through the relaxation dependence of the Rhie–Chow dissipation, the pressure differences being concentrated in the cut cells. The body given as an STL (a 3600-gon) or as a fraction field

gives the accuracy of its geometry route. With two bodies of mixed wall conditions (a no-slip cylinder of radius 1 and a free-slip one of radius 0.7) the forces sum to the driving force times the fluid volume to all printed digits, $0.625915 + 0.202512 = 0.828427$, the check of Proposition 6; the free-slip body carries no viscous force. A four-rank parallel run reproduces the serial drag to all printed digits.

Table 5: Body at rest, no slip unless stated; reference drag 5.660008 (no slip), 2.935529 (free slip).

	steady	transient, ref. Δt	STL body	fraction field	free slip, transient
α , cell by cell	exact	exact	exact	exact	exact
θ , max. diff.	7.6×10^{-11}	7.6×10^{-11}	6.4×10^{-6}	0.11	7.6×10^{-11}
d_{wall} , max. diff.	10^{-11}	10^{-11}	2.8×10^{-6}	2.2×10^{-2}	10^{-11}
u , max. diff. / max u	2.2×10^{-4}	9.4×10^{-6}	2.2×10^{-4}	3.3×10^{-3}	1.7×10^{-5}
p , max. diff. / max p	3.0×10^{-2}	9.6×10^{-4}	3.0×10^{-2}	9.8×10^{-2}	1.9×10^{-4}
drag	5.660383	5.659997	5.660383	5.651222	2.935496

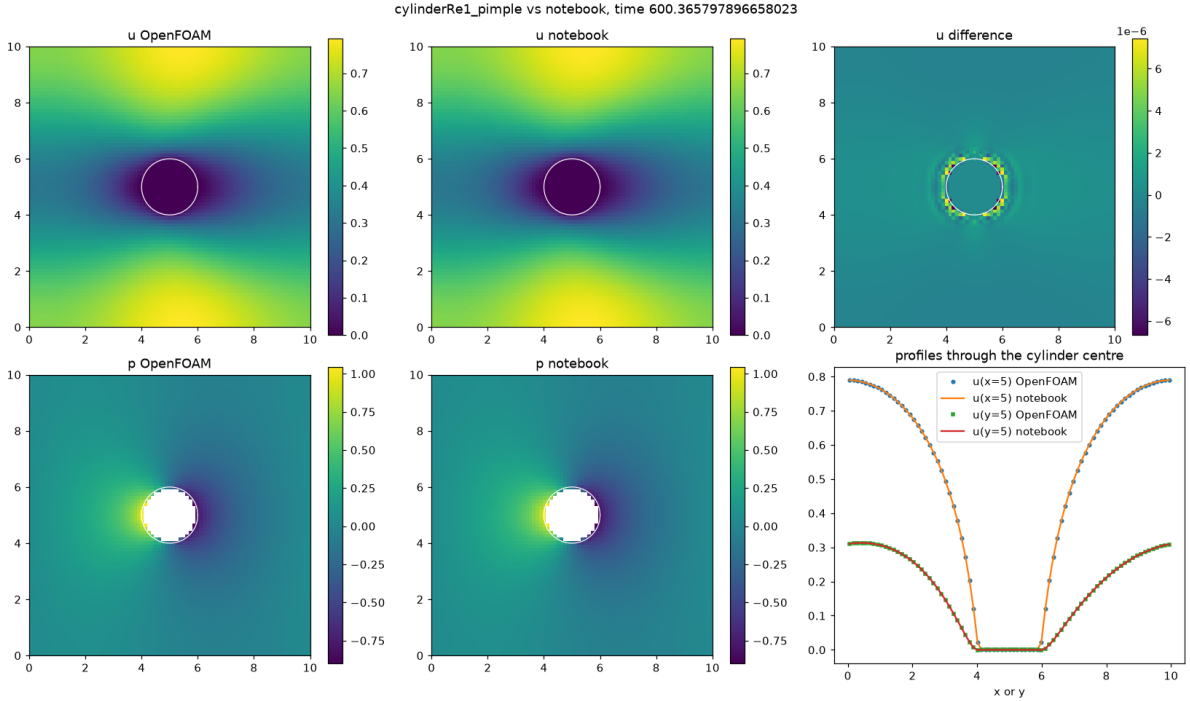


Figure 7: Body at rest, transient solver with the reference’s pseudo-time step: streamwise velocity (top) and pressure (bottom) against the reference, difference field, and profiles through the cylinder centre.

10.2 Moving body: free stream, oscillation, Galilean invariance

The moving cases use $\Delta t = 0.025$, three outer iterations, one corrector, SIMPLEC and no transient Rhie–Chow correction (`ddtCorr no`), as the reference.

Free-stream preservation. The domain, solid included, is filled with $\mathbf{u} = \mathbf{u}_b$ and the body translates at $\mathbf{u}_b$ for five steps, with the forcing targeting $\mathbf{u}_b$ (Proposition 7). The maximum departure from $\mathbf{u}_b$ over the fluid is 0.0 in u and below 2×10^{-15} in v for $u_b = 0.37$ and $u_b = 2.0$ and for one or three outer iterations; the reference gives 5.6×10^{-16} and 3.6×10^{-15} . The test exercises the two-level unsteady term, the mass source, the moving-wall traction, the blanking and the flux re-interpolation simultaneously; a sign error in any of them gives an $O(1)$ result.

Oscillating cylinder. $x_c = 5 + \sin(2\pi \cdot 0.05 t)$, $t \in [0, 10]$, 400 steps, bulk velocity held at zero. The force histories agree to 4.4×10^{-4} after the impulsive first ten steps (7×10^{-5} of the peak

force; the first step, $F = -60.6$, agrees to 0.2%), the driving force to 2.6×10^{-6} , and the final fields to 2×10^{-7} in velocity (Fig. 8). The domain sum of the mass source is below 5×10^{-18} and the continuity error below 2×10^{-16} at every step (Fig. 9); in 25 of the 400 steps one or two cells at the rear of the body have $\alpha_P^n < 0$, without consequence. A four-rank run reproduces the serial forces to 2×10^{-12} .

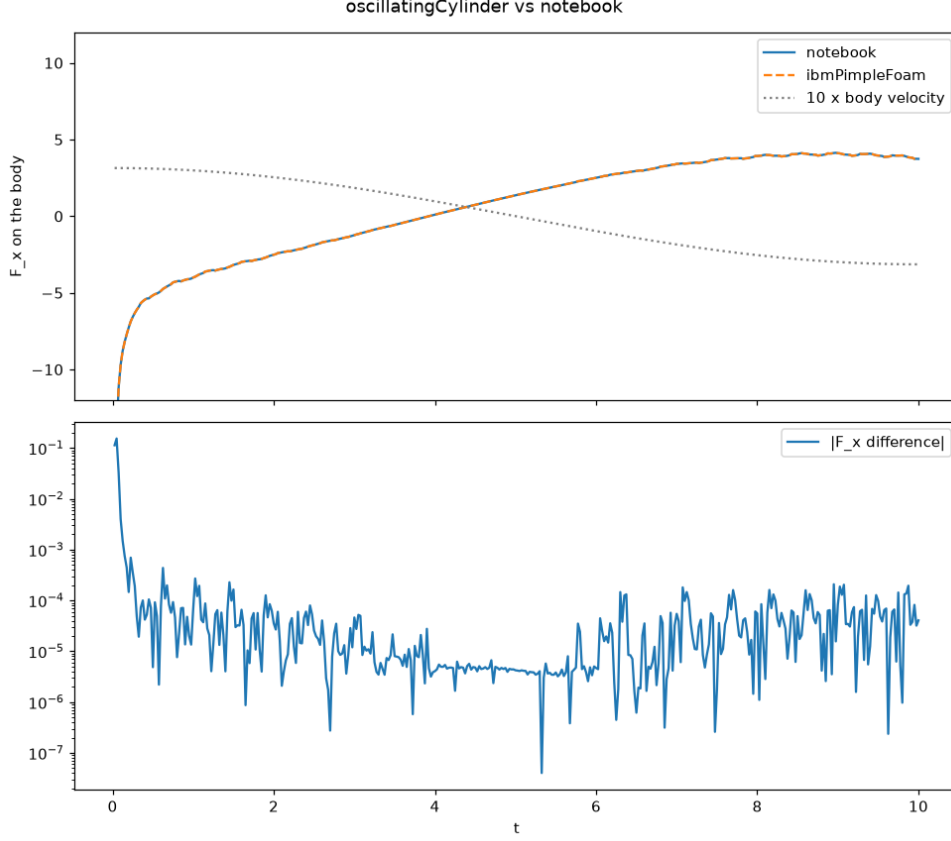


Figure 8: Oscillating cylinder: streamwise force on the body against the reference (top) and their difference (bottom).

Galilean invariance. A body translating at $u_b = 0.5$ through a fluid of zero bulk velocity is the Galilean image of the body at rest in a bulk flow of 0.5, but is computed through entirely different terms. Over $t \in [0, 30]$ the body crosses the periodic seam twice. The mean force over the last 400 steps is -5.69099 with standard deviation 0.0883, against -5.69103 ± 0.0884 for the reference; the steady solver gives $+5.66038$ for the body at rest. The standard deviation (1.5%) is the body sliding from one cut pattern to the next, the characteristic signature of a cut-cell method on a non body-fitted mesh; the 0.5% difference between the moving and the fixed body is not analysed further here (the swept-volume integration is first order in time, and the moving body averages the first-order wall error over all cut patterns while the fixed one sees a single pattern). The histories agree with the reference to 1.3×10^{-3} at every step after the impulsive start (Fig. 10).

This test exposed a subtlety of the seam treatment. The reference, as first written, copied θ across the periodic seam *before* absorbing slivers, so an absorbed cell at $x = L$ closed its face on that side while the same face stayed open for the cell at $x = 0$ (Fig. 2c); while the body front or back crosses the seam its force deviates by up to 2.7% for a few steps. Synchronising the coupled faces with the minimum *after* absorption (Section 3.4) keeps closure exact, the mass-source sum being 10^{-19} throughout; applying the same rule to the reference removes the deviations, and the numbers above are against that reference.

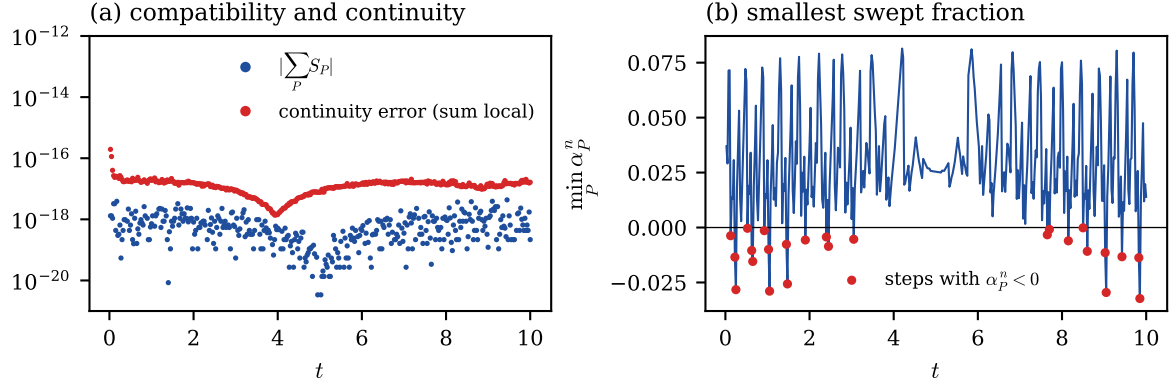


Figure 9: Oscillating cylinder, per-step diagnostics of the solver. (a) Domain sum of the mass source and continuity error (volume-weighted mean of $|\sum_f \phi_f + S_P| \Delta t$ over the correctors). (b) Smallest swept fraction α_P^n ; red dots mark the steps with cells where the step restriction is violated.

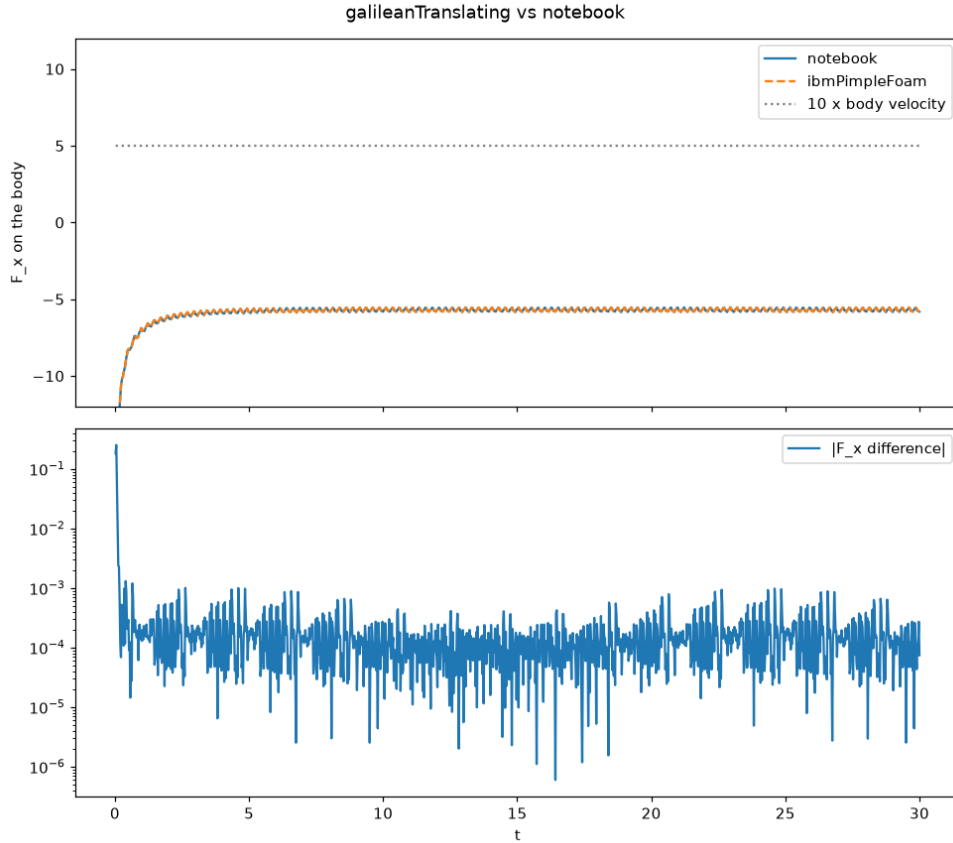


Figure 10: Translating cylinder crossing the periodic seam twice: force history against the reference (top) and their difference (bottom).

10.3 Rotating bodies

Three tutorial cases check the rotational source (24): a cylinder turning in place (`rotatingCylinder`), the same cylinder turning about a point off its axis (`eccentricRotation`) and Taylor–Couette flow (`order/taylorCouette`). At every step the solver logs the domain sum of the source, its largest value and the largest change of the fluid fraction, $\max |\alpha_P^{n+1} - \alpha_P^n|$. The three translating cases above are bit-identical before and after the change, in every field of every write time and in the force histories.

Cylinder turning in place ($R = 1$, $\omega = 1$, centre off the mesh lines, 81^2 cells, $h = 0.123$). With the earlier point evaluation the source was non-zero in all 63 wall cells (largest $|S_P| = 1.2 \times 10^{-3}$ per cell depth) and its domain sum was -1.9×10^{-4} instead of zero, the latter from a single nearly full cell whose evaluation point $\mathbf{x}_P + d_{\text{wall}}\mathbf{n}_w$ lay five units away and was wrapped by the minimum image. With (24) the source vanishes to round-off in 58 of the 63 wall cells; the 5 cells next to removed cells carry $|S_P| \leq 1.3 \times 10^{-3}$ and $|\alpha_P^{n+1} - \alpha_P^n| \leq 0.018$, the flux through their closed faces (Proposition 8); the domain sum is 10^{-18} . That nearly full cell now also receives the right wall velocity for its traction, $|\mathbf{u}_b| = 1.01$ instead of 4.5, the evaluation point being limited to the cell in 16 wall cells (Section 6.3).

Eccentric rotation. The same cylinder turning about a point $(0.3, -0.2)$ off its axis (motion entry `centreOfRotation`) sweeps, uncovers and absorbs cells every step. The domain sum of the source stays below 10^{-16} at every step for sources up to 9×10^{-3} ; a four-rank run, with processor boundaries through the body, reproduces the serial largest source at every step.

Taylor–Couette flow: Section 10.5.

10.4 Passive scalar

A scalar initially zero is injected at unit rate in a disk of radius 0.5 centred at $(5, 8.5)$, away from the path of the body (52 cells, injection rate 0.0978472), with $D = 0.1$ (cell Péclet number about 0.6), and the budget (30) is followed (Table 6, Figs. 11 and 12). With zero-flux walls and a body at rest the content equals the injected amount to 4×10^{-12} after 202 steps and the solid stays exactly at zero: nothing is lost into the solid. With fixed-value walls the amount that left through the wall closes the budget to 6×10^{-11} , and in the steady solver the source rate equals the wall flux rate to 10^{-11} . With a moving body the content differs from the injected amount by the swept-volume defect only, 3×10^{-5} of the content for the oscillating cylinder and 8×10^{-5} for the translating one, the solid stays at the wall value (to 10^{-12}), and the residual budget errors of 10^{-10} are accumulated linear-solver tolerance. With three scalars in one run — one with fixed-value walls, one with zero-flux walls, and one with zero-flux walls and a first-order decay $s = -0.05d$ from a finite-volume option — the first two reproduce the single-scalar runs and the third closes its budget, decay included, to 2×10^{-12} . The flow is unaffected by the scalars.

Table 6: Scalar budget (30) at the end of each run.

case	steps	content	injected	through walls	swept defect	budget error
at rest, zero flux (transient)	202	19.581366	19.581366	0	0	-3.9×10^{-12}
at rest, fixed value	202	11.904949	19.581366	7.676417	0	-5.7×10^{-11}
at rest, fixed value (steady)	1834	16.323010	rate 0.097847	rate 0.097847	—	-1.0×10^{-11}
oscillating, zero flux	400	0.978439	0.978472	0	3.3×10^{-5}	-8.0×10^{-11}
oscillating, fixed value	400	0.968143	0.978472	0.010344	-1.5×10^{-5}	-6.1×10^{-11}
translating, zero flux	1200	2.935650	2.935415	0	-2.3×10^{-4}	-7.5×10^{-10}

10.5 Second-order wall treatment

Operators. The manufactured problems have the fluid inside a disk of radius 1 in a 3×3 periodic box, $N \times N$ cells, $h = 3/N$:

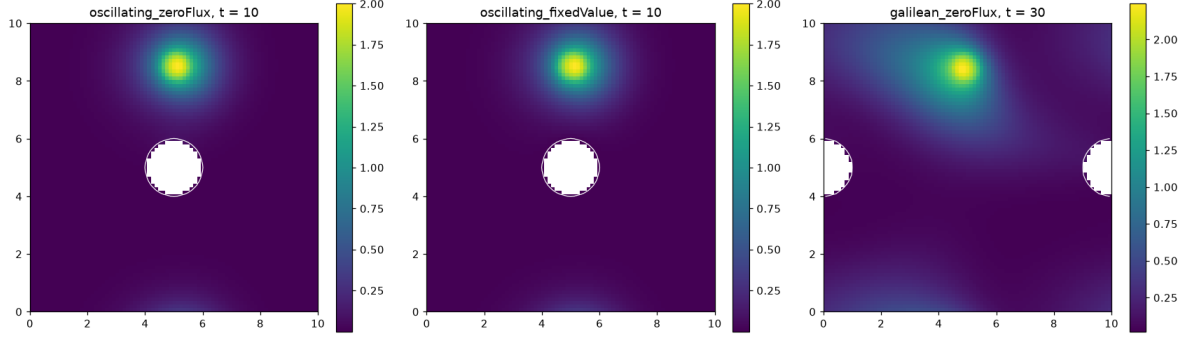


Figure 11: Scalar field for the oscillating cylinder with zero-flux (left) and fixed-value (centre) wall, and for the translating cylinder with zero-flux wall (right); the body is blanked.

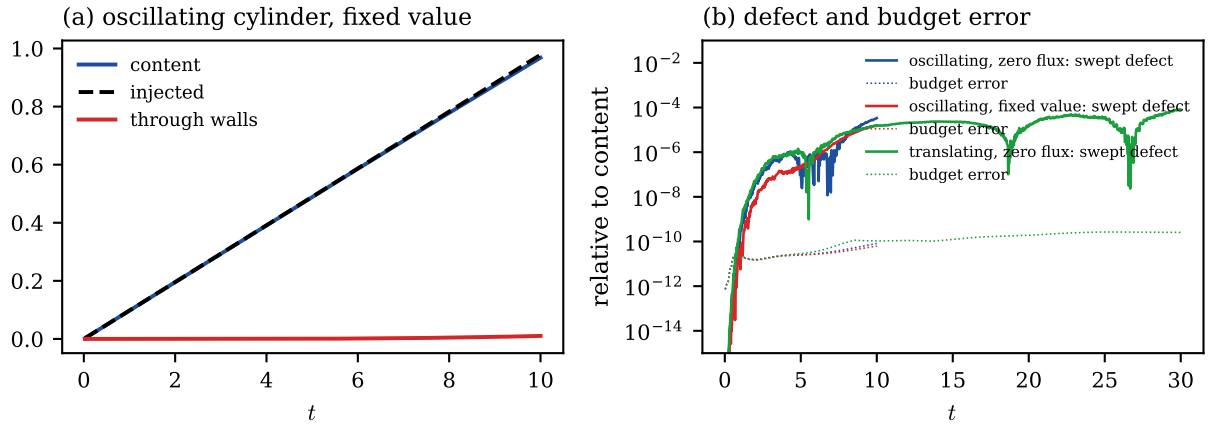


Figure 12: Scalar budget over the moving runs. (a) Content, injected amount and amount that left through the fixed-value wall. (b) Swept-volume defect and budget error, relative to the content: the defect is the only term that does not close, and it stays below 10^{-4} .

test	equation	wall	exact solution
LD	$\nabla^2 w = 1$	$w = 0$	$w = -(R^2 - r^2)/4$
LN	$\nabla^2 w = 16r^2 - 8R^2$	$\partial_n w = 0$	$w = r^4 - 2R^2 r^2$
CD	$\mathbf{u} \cdot \nabla T - \nabla^2 T = q, \mathbf{u} = \Omega(-y, x)$	$T = 0$	$T = (R^2 - r^2)xy$
CN	same, $\Omega = 10$	$\partial_n T = 0$	$T = (R^2 - r^2)^2 xy$

Solid-body rotation is tangential to the wall and divergence-free, so the convection tests exercise the wall treatment with convection without a flow solve; the cell Péclet number is 1.5 at $N = 20$. The prescribed flux is evaluated at the wet-face centroids, errors are α -weighted norms at the fluid centroids, and for the Neumann problems the source is projected onto the compatible space (otherwise the incompatibility created by sliver absorption is deposited at the reference cell, Lemma 1). With the first-order treatment the Dirichlet problems converge at order about one and the Neumann problems at order two; with the second-order treatment all four converge at order two, with errors 100 (LD) and 20 (CD) times smaller at $N = 160$ (Fig. 13). The OpenFOAM operators reproduce an independent Python implementation of the same schemes to the printed digits for the variants both compute. The ablation of Fig. 14, computed with the Python implementation, isolates the ingredients: the centroid wall distance alone (V1/F0) does not raise the order, the centroid-consistent face flux does, a normal probe (V3) adds nothing and the row scaling (V2) is inconsistent.

Pressure gradient. Two further problems exercise the gradient (40) alone on the same geometry, with p given at the fluid centroids: GL, $p = \mathbf{G} \cdot \mathbf{x}$, and GQ, $p = X^2 + \frac{1}{2}XY - 0.8Y^2$ about the disk centre. The relative error of $\nabla_V p / (\alpha V)$ in the wall cells, against $\mathbf{G}$ or $\nabla p(\mathbf{x}_c)$, is

N	20	40	80	160
GL, previous form	0.79	0.76	0.74	0.98
GL, (40)	3.4×10^{-2}	2.5×10^{-2}	1.1×10^{-2}	9.1×10^{-3}
GL, (40), against $V^{\text{geo}} \mathbf{G}$	5.6×10^{-15}	1.6×10^{-14}	2.2×10^{-14}	6.8×10^{-14}
GQ, previous form	0.29	0.41	0.34	0.48
GQ, (40)	3.0×10^{-2}	1.6×10^{-2}	7.8×10^{-3}	4.7×10^{-3}

(L_2 over the wall cells; the first-order form gives 0.53–0.79 for GL.) Against the volume of the geometric polyhedron, obtained from the same operator as $V^{\text{geo}} = [\nabla_V x]_x = [\nabla_V y]_y$, the linear pressure is exact to round-off (Proposition 11); the remaining error against αV is the sub-sampling error of α , relatively large in the smallest cells. The same holds in three dimensions, inside and outside a sphere on 12^3 to 36^3 cells ($\leq 1.2 \times 10^{-13}$, with $\sum_k \mathbf{S}_k = \mathbf{S}_w$ to 2×10^{-11}). For the quadratic pressure the cut-cell gradient converges at first order, as expected of face values that are second order in a one-sided stencil; the previous form did not converge at all.

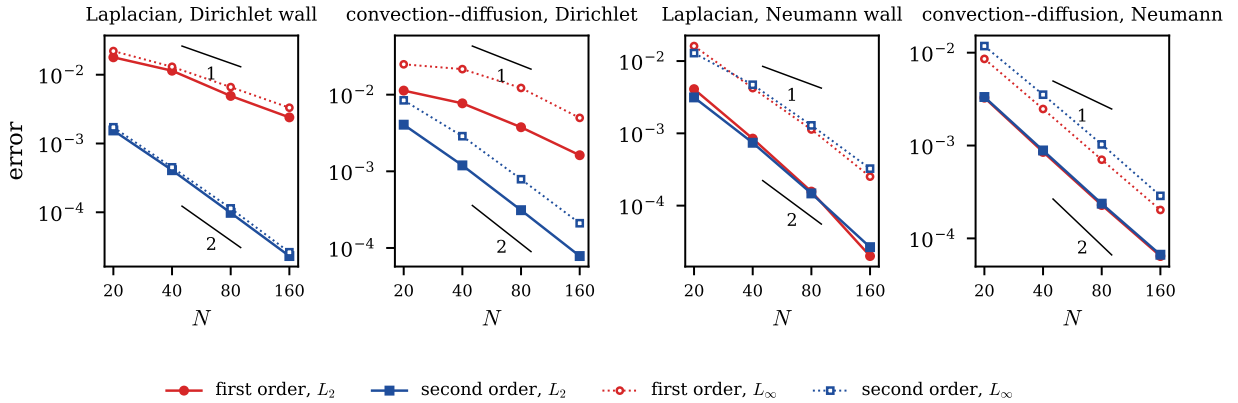


Figure 13: Manufactured problems with the OpenFOAM operators (`ibmOrderTest`): L_2 and L_∞ errors with the first-order treatment (V0/F0) and the second-order treatment (V1/F1); convection uses the face-centre value (C0) in both.

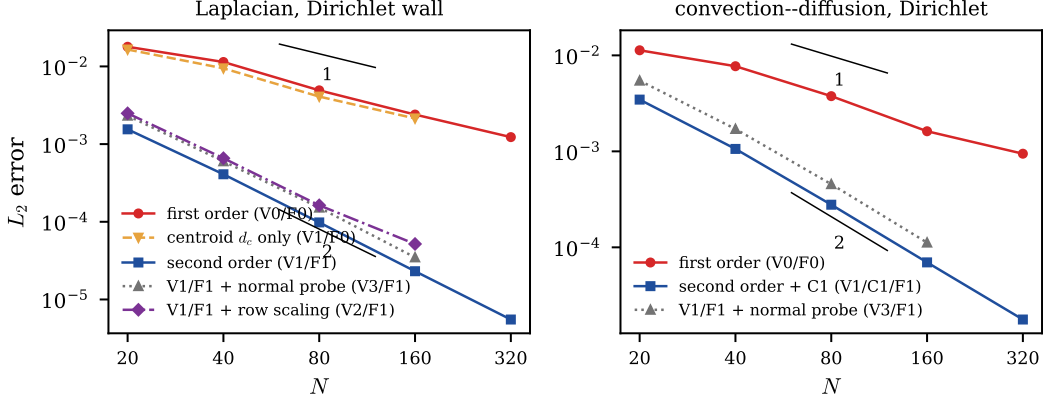


Figure 14: Ablation of the ingredients on the Dirichlet problems (Python implementation, up to $N = 320$): V0/V1 wall distance (7)/(38); F0/F1 face flux of cell centres / (35); V2 row scaling with the cell Laplacian; V3 three-point normal probe; C1 convective value at the wet-face centroid.

Taylor–Couette flow between an inner cylinder of radius 0.5 rotating at $\Omega = 1$ (no slip) and an outer cylinder of radius 1.5 at rest (no slip, fluid inside), both immersed in a 3.2×3.2 box, $\nu = 1$, transient to steady state ($\Delta t = 0.02$, $t = 6$, two outer iterations). The exact solution is $u_\theta = Ar + B/r$, $p = A^2 r^2/2 + 2AB \ln r - B^2/(2r^2)$ with $A = -0.125$, $B = 0.28125$. The first-order runs are evaluated at the cell centres, the second-order runs at the fluid centroids. L_2 errors with the second-order treatment:

N	$10^4 L_2(\mathbf{u})$			$10^3 L_2(p)$		
	40	80	160	40	80	160
before (centroid-point source, previous gradient)	11.6	3.35	1.27	21.7	17.0	13.0
source (24) only	11.7	3.04	0.809	12.0	8.97	4.94
gradient (40) only	11.5	3.60	1.43	22.2	17.2	13.0
both	11.7	3.04	0.812	12.3	9.58	5.13

With both, the velocity converges at order 1.9–2.0 (the largest error, 4.4×10^{-4} at $N = 160$, at order 1.8 against 0.4 before), and the pressure error halves. The source is responsible for all of it; the linearly exact gradient changes the errors by at most 7%, so the remaining pressure error, which converges at order 0.4 then 0.9 and sits in the cut cells (largest 0.19 at $N = 160$), is the small-cell problem of Section 8.6 and not the consistency of the gradient (the row ‘both’ includes the evaluation point limited to the cell, which changes it only at $N = 160$, in the fourth digit). The geometric source with handover (Section 6.3) gives instead $10^4 L_2(\mathbf{u}) = 28.2, 6.32, 8.93$ and $10^3 L_2(p) = 98.7, 40.2, 124$: the velocity loses its second order. Without the second-order treatment the source (24) and the limited evaluation point lower $L_2(p)$ from 2.3×10^{-2} to 1.5×10^{-2} at $N = 160$ and $L_2(\mathbf{u})$ from 2.10×10^{-3} to 2.05×10^{-3} (Fig. 15a).

Hydrostatic balance. Fluid at rest inside an immersed container (an inverted cylinder of radius 1.1) around an immersed obstacle of radius 0.4, under $\mathbf{g} = (0, -1, 0)$ applied to the fluid volume αV , in a periodic box whose boundaries all lie in the solid (tutorials/order/hydrostatic); exact $\mathbf{u} = \mathbf{0}$, $p = -y + \text{const}$. At steady state the largest spurious velocity is, for $N = 40, 80, 160$, 4.0×10^{-4} , 9.9×10^{-5} , 2.5×10^{-5} with the first-order treatment, 2.3×10^{-4} , 6.0×10^{-5} , 1.5×10^{-5} with the previous second-order gradient and 1.1×10^{-4} , 2.6×10^{-5} , 6.4×10^{-6} with (40). The rest is the imbalance between the pressure force $V_P^{\text{geo}} \mathbf{g}$ of a cut cell and the body force $\alpha_P V_P \mathbf{g}$, i.e. the sub-sampling error of α , which also leaves the cut-cell pressure error (2.2×10^{-3} at $N = 160$) slightly above that of the previous gradient (1.6×10^{-3}); a body force weighted with V^{geo} would remove it.

Forces. The drag of the periodic cylinder at $Re_D = 1$ on $N = 41, 81, 161, 321$ (Fig. 15b,c; second order with both fixes, first order unchanged by them) shows the limitation quantitatively. For the no-slip wall the second-order treatment brings the drag much closer to its limit (the

first-order value at $N = 161$ is still 1.3% below the second-order value at $N = 321$) and the total converges at an apparent order of 1.6–1.9; but the components show that this is largely cancellation: the pressure drag (2.783, 2.846, 2.882, 2.902) increases at order 0.8, the viscous drag (3.002, 2.975, 2.951, 2.933) decreases at order 0.2–0.4. The free-slip drag, a pure pressure force, converges at order 0.5–0.7 with both treatments. The linearly exact gradient moves the components by up to 0.7% against the previous second-order form but leaves the no-slip total unchanged to 2×10^{-4} and the orders as they were. The forces are thus limited by the cut-cell pressure and by the $O(\alpha_{\min}h)$ boundary displacement of sliver absorption, not by the operators. (The $N = 321$ runs, and the free-slip runs at $N = 161$, stopped at the iteration limit with residuals between 5×10^{-10} and 3×10^{-8} .)

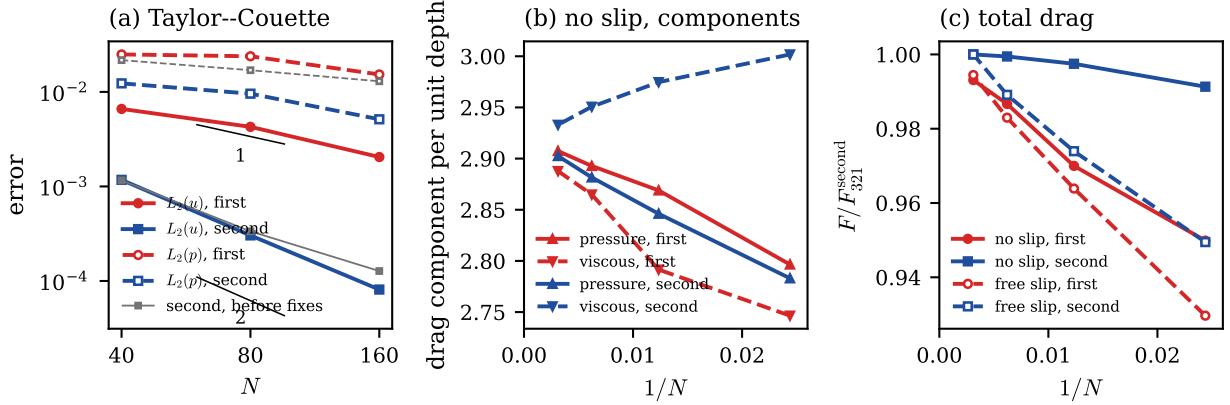


Figure 15: Coupled problems, second order with the rotating-wall source (24) and the gradient (40). (a) Taylor–Couette flow: L_2 errors of velocity and pressure (grey: second order before the two fixes). (b) No-slip cylinder at $Re_D = 1$: pressure and viscous drag. (c) Total drag of the no-slip and free-slip cylinder relative to the second-order value at $N = 321$.

11 Limitations

- *Order.* First order in time (implicit Euler, inherited by the swept-volume integration). First order in space for no-slip walls with the treatment of Section 4, second order for the operators and the bulk velocity with that of Section 8; cut-cell pressure and forces first order in both. A second-order extension in time would need α at three levels and a three-level form of (22).
- *Cut-cell pressure.* The mobility of a cut cell scales with its fluid fraction, so its pressure is poorly determined (Section 8.6); the wall pressure is taken as the cell pressure (integrated over the wall pieces with the least-squares gradient under the second-order treatment, exact for a linear pressure, Section 8.5; the pressure of the cut cells still does not converge). Cell merging or a wall pressure from the full cells is the remedy.
- *Slivers.* Absorption below $\alpha_{\min}$ displaces the boundary by $O(\alpha_{\min}h)$ and removes the fluid volume of the absorbed cell; a cell crossing the threshold hands over no momentum or scalar content. Mass conservation of the flow is unaffected. Cell merging is the clean remedy and would also relax the step restriction. A cut cell whose faces all close (thin features, or after coupled-face synchronisation) while its α stays above $\alpha_{\min}$ is not guarded against. $\alpha_{\min}$ must be positive, otherwise cells with $\alpha = 0$ from a fraction field keep open faces.
- *Sub-cell features.* Faces are classified by their vertices and cells by their vertices, face centres and centre; a feature that passes between them is not seen.
- *Laminar.* Turbulence models are constructed for their effective viscosity only; their own transport equations would need the same treatment as the scalars.

- *Fraction-field bodies.* Reconstructing θ from the 0.5 iso-surface of the point-interpolated fraction smears the wall over one cell (drag -0.16% on the cylinder, θ errors up to 0.11); a per-cell iso-value reproducing each cell's α would bring it to $O(h^2)$. These bodies have no centroids and stay first order.
- *One fluid region.* A single pressure reference; a body splitting the fluid would need one per region.
- *Rotating bodies.* The mass source is the exact flux of the rigid motion through the wall of the post-processed cell, compatible for every rigid motion (Section 6.3). A face closed by sliver absorption belongs to that wall and moves with the body, so a body of revolution turning in place still has a source, and $\alpha^n \neq \alpha^{n+1}$, in the cells next to removed ones; a uniform field is nevertheless preserved and the domain sum is zero. The source is not compatible on faces between cells of two different rotating bodies. In three dimensions the wall of a cut cell is the polyhedral approximation of the surface, so the source of a sphere turning about a diameter is not exactly zero but converges with the geometry. The wall velocity of the no-slip traction is a point value, at (25) limited to the cell.
- *Cost of moving bodies.* The level set is evaluated at all points, face and cell centres every step, and n_s^d samples per cut cell; for three-dimensional triangulated bodies a smaller n_s or a band around the previous position is advisable.

12 Conclusions

The method rests on one geometric statement, the closure identity of a cut cell, and on one choice for moving bodies, the wall flux as the primary quantity. From the closure identity follow exactness for uniform flow, the absence of spurious pressure forces, no-penetration itself, discrete conservation of momentum and exact compatibility of the pressure system; from the flux-first choice follows a discrete geometric conservation law that preserves a free stream to machine precision. The whole boundary treatment adds to the interior scheme two scalar fields, θ_f and α_P , one diagonal coefficient for a no-slip or fixed-value wall, and one source term for a moving wall. Moving the cell values to the fluid centroids, with a centroid-consistent diffusive flux and the centroid wall distance, makes the operators and the velocity second order without ghost values or stencil growth, and integrating the wall pressure over the pieces of the geometric wall makes the pressure gradient exact for a linear pressure; the pressure of the cut cells, and with it the forces, remains the open problem. Implemented in OpenFOAM with explicit operators on an unmodified mesh, it reproduces an independent reference implementation to the level of the linear-solver tolerances for a body at rest, oscillating and translating through a periodic boundary, and transports passive scalars with budgets that close to round-off.